A SIGNED-DIRECTION INVARIANT FOR TRIANGLE TILINGS, AND THE EXCLUSION OF PRIMES CONGRUENT TO 3 MODULO 4

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Abstract. For a triangle T and a triangle ABC, an N-*tiling* is a dissection of ABC into N triangles each congruent to T. Erdős asked which N occur; a folklore conjecture is that no prime $N \equiv 3 \pmod 4$ with $N > 3$ does (the value $N = 3$ occurs). By the classification of Laczkovich and the branch theorems of Beeson, the conjecture reduces to a tile with a $2\pi/3$ angle on a non-equilateral triangle, where a prime count forces ABC isosceles. For that isosceles case Beeson ruled out $N < 36$ and explicitly left open whether N can be prime (the smallest *known* tiling has 2673 tiles); we settle it. We introduce a translation-invariant, signed-direction tiling functional, linear in edge length and hence insensitive to non-edge-to-edge incidences, whose integrality forces, for an isosceles target, that $(c - a - b)/\sqrt{b}$ be an integer for a primitive triple with $c^2 = a^2 + ab + b^2$; an elementary lemma shows this never occurs. With a self-contained reduction of the scalene shapes, and with the base-β branch excluded by the forcing chain of the companion note, this excludes every prime $\equiv 3 \pmod 4$ exceeding 3 *except* the base-β candidates $N = 3f^2 - e^2$, conditional on the cited classification of the remaining branches. Those candidates are exactly the primes $\equiv 11 \pmod{12}$ (Theorem 4), so the exclusion is unconditional on the class $7 \pmod{12}$ — half of all primes $\equiv 3 \pmod 4$ by Dirichlet, and the class containing 19. The class $11 \pmod{12}$ remains open in general; its members are settled individually by exact search, and 42 of them lie below 1000, of which six are so far resolved. Beyond primes, we determine the admissible spectrum of each sporadic $2\pi/3$ branch — for the isosceles target, writing $b = de^2$ with d squarefree, every count has the form $N = dw^2(a+2b)$ with $e \mid w(c-a-b)$ — bounding the realizable set from outside; and a spectrum theorem shows that the counts passing all invariant conditions on the isosceles and F_1 targets are *exactly* Zhang's constructed families, closing the necessary side of those branches; the equilateral criteria likewise reduce to elementary divisor conditions. Combining the spectra with the published branch theorems and an exhaustive exact search of the surviving finite instances, we settle every previously undetermined value up to 80: no triangle can be cut into 14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 51, 55, 56, 57, 60, 63, 69, or 76 congruent triangles (46, 62, 78 likewise, and 79 — a prime — by the main theorem); triangles *can* be cut into 44, 77 (by Beeson's four-component construction), and 80 (a sum of two squares) congruent triangles; the values 66 and 70, previously excluded via theorems of [3] whose proofs we show to be unsound, are re-excluded here by direct exhaustive search of their single surviving instances, so that every $N \leq 80$ is determined. The same method carries the determined initial segment past 80: 86 and 87 are excluded here (the latter by exhausting both of its surviving instances), and 81, 82, 85, 89, 90 are realizable, leaving 84 and 88 under search and 83 excluded only conditionally — it is the first value whose exclusion depends on the base-β branch hypothesis rather than on an N-form, a prime congruence, or an exhaustion. Two of those instances come from branches too sparse to be caught by N-forms: the isosceles base-α shape, which has just three instances in all of $N \leq 90$, and the scalene $2\pi/3$ shape F_4, whose minimal count is exactly 88. The 44-tiling is exhibited explicitly — the isosceles $(16, 16, 22)$ cut into 44 copies of the $(2, 3, 4)$ tile, found by search and verified exactly, to our knowledge the first tiling in the isosceles $3\alpha + 2\beta = \pi$ cases and the smallest known tiling in any incommensurable branch. A collar induction driven by a kernel-verified 22-tile parallelogram certificate realizes $N = 11m^2$ for every $m \geq 2$, and with the branch exclusion at $m = 1$ the instance spectrum of this (target, tile) pair is completely determined: tileable if and only if $m \neq 1$, apparently the first incommensurable pair with no undecided instance. The phenomenon does not persist along the family. For the next member the scale-2 target admits no tiling at all ($N = 104$ on the tile $(3, 8, 9)$, by exhaustive search), so the unit parallelogram is constructible for every member of that family while the seeds of the induction are not, and the instance spectrum depends on the member rather than following a single law. Membership in the tile-count set is shown to be decidable. The arithmetic and combinatorial layer is machine-checked in Lean 4 with Mathlib.

1. Introduction

Let T be a triangle (the *tile*). An N-*tiling* of a triangle ABC is a way of writing ABC as a union of N triangles, each congruent to T, with pairwise disjoint interiors. We do *not* assume the tiling is edge-to-edge: a vertex of one tile may lie in the relative interior of an edge of another. Erdős asked (reported by Soifer [15]) to determine the set of N for which some triangle admits an N-tiling by some tile. Every perfect square occurs (any triangle, by the n^2 subdivision), and Soifer observed that $n^2 + m^2$, $2n^2$, $3n^2$ and $6n^2$ occur as well; in particular $3 = 3 \cdot 1^2$ occurs. The outstanding question is whether the primes $\equiv 3 \pmod 4$ that exceed 3, which are exactly the odd primes that are neither sums of two squares nor of the form $3n^2$, are excluded.

Theorem 1. *Let $N > 3$ be a prime with $N \equiv 3 \pmod 4$ which is* not *a base-β candidate — that is, $N \neq 3f^2 - e^2$ for every coprime pair $1 \leq e < f$. Then N is not a number of congruent triangles into which some triangle can be cut. In particular, since 19 is not of that form, no triangle can be cut into 19 congruent triangles.*

Theorem 2 (Full prime exclusion, conditional)**.** *Assume the complete-corner-wall hypothesis of the companion note* [1, Hyp. 5.1]*. Then no prime $N \equiv 3 \pmod 4$ with $N > 3$ is a number of congruent triangles into which some triangle can be cut.*

Proof. By Theorem 1 and Theorem 4 it remains to exclude the base-β instances, i.e. the primes $\equiv 11 \pmod{12}$, at $m = 1$ for each representation (e, f). Under the stated hypothesis this is the base-β branch theorem of the companion note [1]. The argument there is a finite forcing chain (walk trichotomy; corner, partner and strip forcing; the surplus lattice; the column–filler recursion with its orientation congruences; the feet budget; the T_{mid} corner kill and the pentagon lemma), with the arithmetic components kernel-checked in Lean and the smallest members additionally settled by certified exhaustive refutations. Each representation of a prime is a separate instance and each is excluded. □

Remark 3 (What is unconditional, and what is not)**.** The hypothesis in Theorem 2 is not cosmetic and we flag it plainly. It asserts that in a base-β tiling neither base corner is starved or broken, so that each corner block is a complete wall of whole b-edges. The companion's funnels and its pentagon lemma both consume it, and the companion itself identifies the same statement — that the mismatch segment reaches the base — as the highest-value open step of the branch. It is not a formalization debt: it is an unwritten mathematical step, and until it is supplied, the folklore conjecture is not resolved here.

Unconditionally, this paper proves: no prime $N \equiv 7 \pmod{12}$ is a tile count (such N are not base-β candidates at all, by Theorem 4, so Theorem 1 applies with no exception) — in particular $N = 19$; and the individual exclusions $N = 11, 23, 26, 47, 59, 66, 71, 107$, each by a certified exhaustive search of every surviving instance, which depend on no branch theorem whatever. The class $11 \pmod{12}$ beyond those values, $N = 83$ first among them, is *not* settled unconditionally.

Theorem 4 (The exceptional set is a congruence class)**.** *Let $p > 3$ be prime with $p = 3f^2 - e^2$ for some coprime e, f. Then $p \equiv 11 \pmod{12}$. Conversely every prime $p \equiv 11 \pmod{12}$ is of that form. Hence the base-β candidates among the primes are exactly the primes $\equiv 11 \pmod{12}$. The forward direction is machine-checked (`lean/Mod12.lean`, `base_beta_prime_mod12`), axiom-free beyond `propext`, `Classical.choice` and `Quot.sound`; it is the direction that makes the class $7 \pmod{12}$ unconditional. The converse uses quadratic reciprocity and the class group of discriminant 12 and is not formalised.*

Proof. ($\Rightarrow$) If $3 \mid e$ then $3 \mid p$, impossible for a prime $p > 3$; so $e^2 \equiv 1 \pmod 3$ and $p \equiv -1 \equiv 2 \pmod 3$. Next, e, f are not both even (coprimality) and not both odd, since then $p = 3f^2 - e^2 \equiv 3 - 1 = 2 \pmod 4$ would be an even prime exceeding 3; in either remaining case $p \equiv 3 \pmod 4$. Combining, $p \equiv 11 \pmod{12}$. ($\Leftarrow$) For $p \equiv 11 \pmod{12}$ we have $p \equiv 3 \pmod 4$ and $p \equiv 2 \pmod 3$, so by reciprocity $\left(\frac{3}{p}\right) = 1$ and p splits in $\mathbb{Q}(\sqrt{3})$. The discriminant-12 form class group has order two, represented by $x^2 - 3y^2$ and $3y^2 - x^2$; the first forces $p \equiv 1 \pmod 3$ and the second $p \equiv 2 \pmod 3$, so p is represented by $3y^2 - x^2$. Primitivity is automatic for prime p. $\square$

Theorem 5 (The first complete member)**.** *Let the tile be $(2,3,4)$ and the target the isosceles triangle with base angles β (the base-β member $(e,f) = (1,2)$, instances $N = 11m^2$). Then the instance is tileable if and only if $m \neq 1$: every $N = 11m^2$ with $m \geq 2$ is realized, and $N = 11$ is excluded. This is, to our knowledge, the first (target, tile) pair with incommensurable angles whose instance spectrum is completely determined.*

Proof. Exclusion at $m = 1$ is the base-β branch theorem [1]. Realizability is the collar induction of the companion note: the collar of Δ_m (the bottom two rows of the m-fold cell subdivision) decomposes into $m-2$ sheared parallelogram columns and one scale-2 corner triangle, all interfaces full straight segments; each column is two translates of the unit parallelogram. The three seeds are found by exhaustive search and kernel-verified with exact $\mathbb{Z}[\sqrt{15}]$ arithmetic and zero axioms (`Tiling44.lean`, `Tiling99.lean`, `PgramTiling22.lean`: congruence, containment, an explicit separating line per tile pair, and the exact area identity), and the two-step induction from bases $m = 2, 3$ is itself kernel-checked (`Collar.lean`). See the companion note [1] for the full statement and proof. $\square$

Remark 6 (The spectrum is member-dependent)**.** Theorem 5 does not extend along the base-β family. For $(e,f) = (1,3)$ — tile $(3,8,9)$, $N_1 = 26$ — the scale-2 target admits no tiling: $N = 104$ is excluded by exhaustive search (43 541 916 nodes). The two ingredients of the induction therefore behave differently along the family: the unit parallelogram is constructible for every member with $e = 1$ (companion note, Theorem on the unit parallelogram; verified for $f = 2, \ldots, 12$), while the seeds Δ_2, Δ_3 that the collar needs are not. Every attempt to build the seed at $f = 3$ fails for the same reason, namely that there is nothing to build: the cevian piece, the symmetric middle piece, and the maximal interiors of both natural lattices all return NO TILING, and a seeding by unit parallelograms is impossible by counting. Consistently with this, $(1,2)$ is the unique member of its family with $\beta < \pi/3$. What the spectrum of $(1,3)$ is remains open: since Δ_4 would require $\Delta_1 + \Delta_3$ and Δ_5 would require $\Delta_2 + \Delta_3$, both excluded, the set reachable from Δ_3 alone is exactly the multiples of 3.

Corollary 7 (Theorem 1, congruence form)**.** *No prime $p \equiv 7 \pmod{12}$ is a number of congruent triangles into which a triangle can be cut. Equivalently: a prime $p > 3$ with $p \equiv 3 \pmod 4$ is excluded unconditionally unless $p \equiv 11 \pmod{12}$.*

Proof. A prime $p > 3$ with $p \equiv 3 \pmod 4$ satisfies $p \equiv 7$ or $p \equiv 11 \pmod{12}$, the class $3 \pmod{12}$ forcing $3 \mid p$. If $p \equiv 7 \pmod{12}$ then by Theorem 4 it is not a base-β candidate, so Theorem 1 applies with no exception. $\square$

Remark 8 (Why the forms, and what the reformulation buys)**.** Every tile count in the classification is a value of a binary quadratic form: $e^2 + f^2$ (discriminant -4) on the commensurable branch, the Eisenstein norm $a^2 + ab + b^2$ (discriminant -3) for a $2\pi/3$ tile, $3f^2 - e^2$ (discriminant 12) and $2f^2 - e^2$ (discriminant 8) on the $3\alpha + 2\beta$ branch. The classical exclusion of primes $\equiv 3 \pmod 4$ on the commensurable branch is exactly Fermat's two-squares theorem for the first form; Theorem 4 is its analogue for the third. Two consequences. First, the exception in Theorem 1 is not a Diophantine side condition but a single arithmetic progression, and by Dirichlet it carries half of the primes $\equiv 3 \pmod 4$: the theorem is unconditional on the other half. Second, $19 \equiv 7 \pmod{12}$, so the value in Erdős's question lies in the unconditional class. The forward implication is machine-checked (`basebeta_prime_mod_twelve`, `BaseBetaMod12.lean`); it is the direction the corollary uses. The converse was checked for all 109 primes $\equiv 11 \pmod{12}$ below 3000 with no exception.

Remark 9 (The quoted form of [3, Lem. 6] is false)**.** That lemma is usually quoted as "every side of ABC carries at least two c-edges". In that form it is false. Take the tile $(2,3,4)$ and let $ABC=(4,6,8)$, its double; the four-fold midpoint subdivision is a genuine tiling by four congruent copies, and its sides decompose as b^2, c^2, a^2 — two of the three carry *no* c-edge. Here ABC is similar to the tile, a case treated separately in [3], so the true statement must carry hypotheses that the usual quotation omits. We therefore do not use it: Proposition 30 now invokes only the γ-trap (at least *one* c-edge per side), which is proved here from the vertex figures and machine-checked, and which suffices because the base length identity gives $Y \le (g-M)b$ unconditionally — a single c-edge already contradicts. An exhaustive check over all 11,338 admissible pairs (N,M) with $N<2500$ prime found *no* decomposition of the base containing even one c-edge, so the margin is not delicate; the citation [3, Lem. 45(iii)] is likewise not needed for the conclusion.

Remark 10 (The scope of Theorem 1)**.** The base-β exception is essential. The base-β candidates are exactly the $N=3f^2-e^2$ with $\gcd(e,f)=1$, $1\le e<f$. Those with e,f of opposite parity are $\equiv 3 \pmod 4$; those with e,f both odd are $\equiv 2 \pmod 4$, hence even, so among the *primes* only the former occur — and by Theorem 4 they are exactly the primes $\equiv 11 \pmod{12}$. (Composite candidates of either parity occur and appear in Table 1: 26, 66 and 74 are $\equiv 2 \pmod 4$.) The smallest prime candidates are

$$11,\ 23,\ 47,\ 59,\ 71,\ 83,\ 107,\ 131,\ 167,\ 179,\ 191,\ \ldots$$

For each, the tile $(ef,\ f^2-e^2,\ f^2)$ and the target $(f^3, f^3, e(3f^2-e^2))$ pass every *sound* necessary condition (Remark 13); the only published exclusion is [3, Thm. 14], whose $g \mid M$ is not merely unproven but **false** (Remark 21), so we cannot invoke it, and no general replacement is known [1]. The theorem therefore cannot drop the exception without appealing to that false citation. The candidates are settled one at a time by exhaustive search (Table 1): $11, 23, 47, 59, 71, 107$ are excluded and 83 remains under search. Thus Theorem 1 is unconditional for every prime $N\equiv 3 \pmod 4$, $N>3$, outside the sequence above, and for $11, 23, 47, 71, 107$ besides. The principal consequence is unaffected: $19 \ne 3f^2-e^2$, and the two frontier claims of Section 7 rest only on candidates that have been searched.

The restriction $N>3$ is necessary: an equilateral triangle joined to its centre splits into 3 congruent triangles, so $N=3$ occurs.

The proof rests on Laczkovich's classification of triangle tilings [11, 12], organized by Beeson across a series of papers [2, 3, 4, 5, 6]. For a prime number of tiles every branch of the classification is excluded by an existing theorem, with one exception: a tile having a $2\pi/3$ angle, tiling a non-equilateral triangle. We treat that branch.

Zhang [9] classifies the non-equilateral $2\pi/3$ shapes into six sporadic similarity types (Section 2 below), constructs families of tilings for each, and conjectures that the constructed N, all composite, are the only ones; the present results prove the no-prime half of that conjecture for the isosceles type, and reduce the scalene types directly. For the isosceles type Beeson [4] proved that no tiling exists with $N<36$ and *explicitly left open whether N can be prime*; the smallest tiling presently known has 2673 tiles (Herdt, reported in [4]). The invariant below settles the question, excluding *every* prime. The adjacent Erdős Problem 633, on tilings into a square number of triangles, was recently settled by Beeson, Laczkovich and Zhang [10].

Two reductions prepare it. First, such a tile has commensurable sides, so after scaling we may take it integer-sided with $c^2=a^2+ab+b^2$; for the *isosceles* targets to which the theorem reduces, this is Beeson's *accepted* isosceles paper ([4, Thm. 12.5]: any isosceles triangle tiled by a $2\pi/3$-tile with α/π irrational forces the tile rational), so that citation covers the isosceles targets; but the reduction to an isosceles target (Section 3) already works with an integer-sided tile, so the rationality input actually used in that order is the general-target theorem of Beeson–Zhang, a preprint — we record this plainly rather than claim a refereed input; the general-tile rationality of Beeson–Zhang [8] gives the same uniformly. Second, an elementary area-and-incidence argument shows that a *prime* number of such tiles forces ABC to be *isosceles* (Section 3).

The main construction is in Section 4. The natural parity argument for a $2\pi/3$ tile (a black/white colouring of the tiles) fails, because a vertex of the tiling may be surrounded by an even number of tiles, so no consistent two-colouring exists. We replace it by a functional Φ that assigns to a directed edge of direction $\theta = j\frac{\pi}{3} + k\alpha$ the weight (length) $\cdot (-1)^j$. Two features make it work: $\Phi(\theta+\pi) = -\Phi(\theta)$, so interior edges cancel; and the weight is *linear in edge length*, so the cancellation is unaffected by non-edge-to-edge incidences, where one long edge meets several shorter ones summing to it. Summing over the tiles, Φ of the boundary equals an integer multiple of a fixed tile-value $c + a - b$. For an isosceles target this forces $(c - a - b)/\sqrt{b} \in \mathbb{Z}$, which never holds (Section 5). Assembling the branches proves Theorem 1 (Section 6).

The combinatorial claims are verified by the scripts in the accompanying repository, and the entire arithmetic layer, including Lemma 41 and the reduction of Section 3, is formalized and machine-checked in Lean 4 with Mathlib; see Section 8.

2. Preliminaries

The classification. Fix a prime $N = p > 3$ with $p \equiv 3 \pmod 4$. By Laczkovich [11, 12], every N-tiling of a triangle falls into one of finitely many types of pairs (ABC, T). The relevant dichotomy, in Beeson's organization [5], is: either the angles of T are pairwise commensurable with π, or they are not. If they are commensurable, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [5, Thm. 3 and Table 3]; but $p > 3$ with $p \equiv 3 \pmod 4$ is a square only if p is one (it is not), a sum of two squares only if $p \not\equiv 3 \pmod 4$, and $2n^2, 3n^2, 6n^2$ only if p is even or $3 \mid p$ (here p is odd and $p > 3$), so p is none of these (it is precisely for $p = 3 = 3 \cdot 1^2$ that this step would fail). If the angles are not commensurable, then, after the right-angle and similar-tile cases, which give N a sum of two squares or $N = n^2$ [2, 14], the tile has a special angle and falls into one of: $3\alpha + 2\beta = \pi$, for which N is not prime (originally [3]; see Remark 11 and Propositions 30 and 12 — the arithmetic core is machine-checked in Lean for *four* of the five target shapes, with our own proofs replacing Beeson's Theorems 18 and 20, whose printed arguments are unsound; the fifth, the base-β isosceles target, is the exception, treated in Remark 13); an isosceles tile with $\gamma = 2\alpha$, for which N is not squarefree [4, Thm. 11.7]; or the tile has an angle equal to $\pi/3$ or $2\pi/3$. The case of a $\pi/3$ or $2\pi/3$ tile on an *equilateral* triangle gives N not prime [6]. What remains for a prime N is a tile with a $\pi/3$ or $2\pi/3$ angle on a *non-equilateral* triangle. The vertex enumeration of Section 2 shows the $\pi/3$ case admits only equilateral or tile-similar ABC, already covered; so the sole remaining branch is a $2\pi/3$ tile, which we now fix.

Remark 11 (The $3\alpha + 2\beta = \pi$ prime exclusion)**.** The step above invokes [3] for the claim that a $3\alpha + 2\beta = \pi$ tiling has N composite. For the two scalene targets $(2\alpha, \beta, \alpha+\beta)$ and $(2\alpha, \alpha, 2\beta)$ this is Beeson's Theorems 8 and 12, whose arithmetic cores we machine-check in Lean (`lean/Beeson3NotPrime.lean`): the first tiling equation $N + M^2 = 2K^2$ with $K \mid N$ admits no odd prime, and the second forces $N = (2f^2-e^2)(3f^2-e^2)k^2$ — a product of two factors exceeding 1, hence composite. The isosceles target with base angles $\alpha+\beta$ is Beeson's Theorem 18, but its published proof is *unsound*: under Beeson's scaling $a = N - M^2,\ c = N + M^2,\ b = c - a^2/c$ one has $bc = c^2 - a^2 = 4NM^2$, so $c^2 b = 4NM^2(N + M^2) \equiv 0 \pmod N$ *identically* (no primality used), not the $M^4(M^2 + 1)$ his argument needs — a dropped factor of a; the residue is already false on genuine composite tilings ($N = 48$, $M = 4$ gives $c^2b \equiv 0$ while $M^4(M^2+1) \equiv 32$), so the step forcing $N = M^2 + 1$ is vacuous. The conclusion is nonetheless correct, by the base-length obstruction of Proposition 30.

Two further defects of [3] affect the base-α target. First, the squarefree assertion of its Lemma 8 ("$g = \gcd(a, c)$ is squarefree") is *false*: the tile $(4, 15, 16)$ — a genuine $3\alpha + 2\beta$ tile, $e = 1$, $f = 4$ — has $g = 4$ (its proof breaks at the inference $p^2 \mid a \Rightarrow p^2 \mid \hat{a}$); the parametrized family $(ef,\ f^2 - e^2,\ f^2)$ realizes *every* f coprime to e, squarefree or not. Second, Theorem 19's claim $g \mid M$ rests on that lemma *and* on the assertion that $bc^3(a + c)$ is not divisible by g^5 — but with $c = g^2$ it carries g^7 identically, so the argument collapses, and with it the proof of the base-α prime exclusion (Theorem 20). Proposition 12 repairs the latter without any appeal to $g \mid M$.

Proposition 12 (Base-α prime exclusion, replacing [3, Thm. 20])**.** *No isosceles triangle with base angles α, where $3\alpha + 2\beta = \pi$ and $\alpha \notin \mathbb{Q}\pi$, is N-tiled for N prime.*

Proof. Write $s = e/f$ in lowest terms, so the tile is $(ef,\ f^2 - e^2,\ f^2)$. Two necessary conditions suffice: the tiling equation $N(f-e)(2f+e)^2 = M^2(f+e)(2f^2-e^2)$ (the sound half of [3, Thm. 19]) and the integrality of the equal side X, where $X^2 = Nbc/(2-s^2)$. Substituting the tiling equation into X^2 gives the exact value $X = M(f+e)f^2/(2f+e)$; as $\gcd(2f+e, f) = \gcd(2f+e, f+e) = 1$, integrality forces $(2f + e) \mid M$. Writing $M = (2f + e)m$ reduces the tiling equation to

$$N(f - e) \ = \ m^2\,(f + e)\,(2f^2 - e^2).$$

Put $d = f - e$ and $Q = 2f^2 - e^2 = e^2 + 4ed + 2d^2$. Then $\gcd(d, Q) = 1$ (a common prime would divide e^2, hence e, hence $\gcd(e, f)$), so the minimal prime factor of $Q \geq 7$ divides N and equals N; writing $Q = NQ_1$ and cancelling N leaves $d = m^2(f+e)Q_1 \geq f+e > f-e = d$, a contradiction. Both arithmetic pillars are machine-checked in Lean (`lean/IsoAlphaPrime.lean`, axiom-clean: `isoalpha_X_forces`, `isoalpha_not_prime`). ∎

Remark 13 (The base-β target)**.** The fifth target, isosceles with base angles β, is the one shape where we cannot replace Beeson's argument with a sound one. Beeson's Theorem 14 also asserts $g \mid M$, by a proof of the same broken shape as Theorem 19's (it claims abc^5M^2 is not divisible by g^7, but with $c = g^2$ it carries g^{11}). And here the defect *bites*: the sound necessary conditions do not exclude prime N. With $s = e/f$ in lowest terms the tile is $(ef,\ f^2 - e^2,\ f^2)$, the equal side is $X = f^3M/(f + e)$ and the base is $Y = eM(3f^2 - e^2)/(f + e)$; integrality of X forces $(f + e) \mid M$, and writing $M = (f + e)m$ the Theorem-14 equation $N(e + f)^2 = M^2(3f^2 - e^2)$ collapses to

$$N = m^2\,(3f^2 - e^2),$$

which *is* prime for $m = 1$ and $3f^2 - e^2$ prime (e.g. $N = 11, 23, 47, 59, 71, \ldots$, always $\equiv 3 \pmod 4$). Each such N passes every sound necessary condition ($X, Y \in \mathbb{Z}$, parity, range) and is killed only by the unsound $g \mid M$. Unlike the base-α case (Proposition 12), no arithmetic contradiction follows; *a general proof that these candidates do not tile is open for* $e \geq 2$, and to our knowledge is not supplied by a correct argument in the literature. For the subfamily $e = 1$ the companion note [1] now proves the exclusion outright (its Theorem on the $e = 1$ family, for all even f): this settles by argument the candidates $11, 47, 107, 191, 431, \ldots$ — every prime $p = 3f^2 - 1$ with no second base-β representation — and leaves $e \geq 2$ as the open residue. This assessment includes the recent preprint [7] (21 July 2026), which asserts that no isosceles triangle admits a prime tiling: its Theorem 1 disposes of the whole $3\alpha + 2\beta = \pi$ branch by citation to [3] ("by considering the possible shapes of T one by one"), and for the present shape the argument of [3] is Theorem 14, refuted above by the 99-tiling (Remark 21). The genuinely new content of [7] is the isosceles target with a $2\pi/3$-angle tile, proved there by a short-relation argument (its Theorems 8 and 10); that case is Theorem 42 of the present paper, obtained here by the signed-direction functional. The two proofs are independent and the methods are disjoint; the present proof, together with the repository holding it, was made public on 27 June 2026 and communicated to the author of [7] the same day. For the finitely many instances that bear on the values settled in this paper we resort to the direct corner-anchored search. An exhaustive audit of $N \leq 80$ — computing the equal side X and base Y exactly and retaining every candidate that passes all *sound* necessary conditions ($X, Y \in \mathbb{Z}$, parity, range) — shows the affected set is precisely

$$\text{primes } \{11, 23, 47, 59, 71\} \text{ (base-}\beta) \quad \text{and} \quad \text{composites } \{21, 70\} \text{ (base-}\alpha),\ \{39, 66\} \text{ (base-}\beta).$$

The engine settles them one at a time; all searches are exact and exhaustive:

N	branch	tile	target	nodes
11	base-β	$(2,3,4)$	$(8,8,11)$	135
21	base-α	$(2,3,4)$	$(12,12,21)$	13 659
23	base-β	$(6,5,9)$	$(27,27,46)$	19 677
47	base-β	$(4,15,16)$	$(64,64,47)$	12 440
39	base-β	$(12,7,16)$	$(64,64,117)$	825 724
71	base-β	$(10,21,25)$	$(125,125,142)$	553 417

each returning EXHAUSTED, NO TILING. The remaining three — $N = 59$ (base-β, $(20,9,25)$ on $(125,125,236)$), $N = 66$ (base-β, $(15,16,25)$ on $(125,125,198)$) and $N = 70$ (base-α, Remark 70) — are still searching, and their values are treated as pending. For the base-β target the $3\alpha + 2\beta$ prime argument rests on either [3, Thm. 14] (whose printed proof is unsound) or a per-value search, *not* on a machine-checked general theorem; this is the one remaining gap in the prime dichotomy. The companion note [1] removes $N = 11$ unconditionally, Remark 14 isolates the $e = 1$, $f \geq 3$ structure, and Remark 20 explains why the method [3] used cannot repair the rest.

Remark 14 (The $e = 1$, $f \geq 3$ cases are *not* settled)**.** For $f \geq 3$ the composition $B_1 = (a^f, b, c)$ survives every citation-free constraint above, and it is then the *unique* possible base. Two consequences follow. First, B_1 carries exactly *one* c-edge; so [3, Lem. 6] ("every side carries at least two c-edges") removes it at a stroke — but that citation is doing *all* of the work, and any tiling of this family would *refute* Lemma 6 rather than be excluded by it, so invoking it is tantamount to assuming the conclusion. Given that [3] is the source whose Lemma 8 we show false, whose Theorems 18–20 we show unsound, and whose Theorem 14 we show to be outright *false* (Remark 21), we do not invoke it. Second, our own forcing argument targets B_0, which cannot occur in any case, so it says nothing about $f \geq 3$. We therefore record $N = 26, 47, 74, 107, 146, 191, \ldots$ as *open* in this branch, with $N = 26, 47, 74$ independently excluded by exhaustive search.

What survives citation-free is a sharp structure theorem: *if* an $e = 1$, $m = 1$, $f \geq 3$ base-β tiling exists, then $q_{\text{equal}} = 0$, its base is exactly (a^f, b, c) with $E_1 = E_k = a$ and the c-edge interior, both equal sides begin and end with c, and every base junction is a $(1,1,1)$-vertex.

The next theorem replaces the $e = 1$ case analysis by a classification valid for *every* e, and shows why $m = 1$ — the prime case — is the rigid one. It is machine-checked (`BaseBetaWalks.lean`, axiom-clean).

Proposition 15 (Unsplittability, and the rigidity of the thick regime)**.** *For every coprime $1 \leq e < f$, neither $a = ef$ nor $b = f^2 - e^2$ is a sum of two or more tile edges; and $c = f^2$ is such a sum if and only if $e = 1$, in which case uniquely $c = a^f$. Consequently* in the entire open thick regime, where $e \geq 2$, no tile edge splits at all.

Proof. Reduce $n_a\, ef + n_b(f^2 - e^2) + n_c f^2 = L$ modulo f. For $L = a$: $f \mid n_b e^2$, so $f \mid n_b$ by $\gcd(e, f) = 1$; $n_b \geq f$ overshoots since $fB > ef$ (as $B = f^2 - e^2 \geq 2e + 1 > e$), so $n_b = 0$, and cancelling f leaves $n_a e + n_c f = e$, forcing $n_c = 0$ and $n_a = 1$. For $L = b$: $f \mid (n_b - 1)e^2$, so $n_b \equiv 1$; $n_b \geq 1 + f$ overshoots, so $n_b = 1$ and the rest vanishes, giving $n_a = n_c = 0$. Either way the total is 1, never ≥ 2. For $L = c$: $f \mid n_b$ forces $n_b = 0$ as before, and $n_a e = f(1 - n_c)$ gives $n_c = 0$ and $e \mid f$, i.e. $e = 1$, $n_a = f$. □

Remark 16 (The golden ratio)**.** Two consequences. First, Proposition 15 is exactly the hypothesis the corner parallelogram needs, so Proposition 28 holds with *no* proviso. Second, $e = 1$ — which both genuine tilings realize — is the only case in which an edge splits at all, while the open $e \geq 2$ regime is rigid. The comparison of a with b must be made with care: $b > a \iff f^2 - ef - e^2 > 0 \iff f/e > \varphi$, the golden ratio. The thick regime straddles that threshold: $N = 66$ $(3, 5)$ and $N = 131$ $(4, 7)$ are super-golden, while $N = 23$ $(2, 3)$, $N = 59$ $(4, 5)$, $N = 83$ $(5, 6)$ and $N = 167$ $(5, 8)$ have $b < a$. Proposition 15 above assumes no inequality between a and b.

Remark 17 (Validation on the 44-tiling)**.** On the genuine 44-tiling ($f^2 = 4 > 2e^2 = 2$, so $t = 0$) the piercer spans exactly $[3, 6]$, starting at the T-junction, with corner α there to numerical precision, as forced; the arithmetic core is machine-checked (`pre_pierce_dichotomy`). The rung also has a computational consequence: the apex neighbourhood is rigid (three forced α-tiles, a forced mismatch ray, a forced T-junction, a forced piercer start), while the base admits several walks. Since the corner-anchored search anchors at the lowest vertex, orienting the target apex-down (a reflection, under which verdicts are preserved) places the rigid chain at the root of the search tree: on $N = 23$ the apex-down search exhausts in 3,954 nodes against 19,677, whereas on the thin member $N = 47$, where the base is the rigid side, the apex-down orientation is slower. The thick searches $N = 59, 66$ are run apex-down.

Remark 18 (A parity argument that fails, and why)**.** It is tempting to argue as follows. Every tile has exactly one b-edge; by Proposition 15 the edge b is never a sum of two or more tile edges; so an interior b-edge is matched by exactly one further b-edge, interior b-edges pair the tiles two at a time, and $N \equiv Q_\partial \pmod 2$ with Q_∂ the number of boundary b-edges. We record that this is *false*, because the middle step does not follow.

Unsplittability forbids an edge from being *exactly partitioned* by two or more tile edges. It does not forbid a *straddle*: a longer edge may contain a b-edge in its interior and overhang it at one or both ends, and then the b-edge has no partner. Indeed the apex-mismatch configuration of [1] is built on precisely such a straddle. The genuine tilings confirm the failure: in the 44-tiling six b-edges belong to a single tile although *no* b-edge lies on the boundary, and the 99-tiling has thirty-three matched pairs, eleven boundary b-edges and twenty-two interior straddles. The correct relation is

$$N = 2 \cdot \#\{\text{matched pairs}\} + \#\{\text{boundary}\} + \#\{\text{straddled}\},$$

so the parity conclusion needs the straddle count to be even, which nothing supplies. (It happens to be even, 6 and 22, in these two tilings, which is why the conclusion survives there.)

The same objection defeats the analogous arguments for the a- and c-edges. We state this explicitly because the argument is natural, its conclusion is correct on every example we can test, and it is nevertheless not a proof.

Remark 19 (What the alignment pins, uniformly)**.** For every open thick member the mismatch ray is now completely determined out to length $f \cdot b$: its far wall is b^f, terminating transverse to the opposite side, and $V = f^2$ falls strictly inside the $\lceil f^2/b \rceil$-th b-edge. Explicitly (all exact): $N = 23$ far side $b^3 = 5^3$, $V = 9 \in (5, 10)$; $N = 59$ $b^5 = 9^5$, $V = 25 \in (18, 27)$; $N = 66$ $b^5 = 16^5$,

$V = 25 \in (16, 32)$; $N = 83\, b^6 = 11^6$, $V = 36 \in (33, 44)$. This pins an extended *interior* structure of a hypothetical thick tiling, uniformly in (e, f); it serves as a root prune for the apex-down search; the forced b^f wall must land on the base at coordinate $e \cdot f^2$, whose b-count is bounded (`base_b_bound`). It does not, on its own, close any member.

Remark 20 (Why the colouring method cannot close the rest)**.** The residual gap ($e \geq 2$) is not merely unproved by [3, Thm. 14]: it is *unprovable by that method*. Let $\eta = e^{i(\omega+\pi/2)}$, so that $f\eta^2 + e\eta + f = 0$ and every edge direction of every such tiling lies in $\langle \eta, -1\rangle$. Any T-junction-proof invariant of the shape $\sum_{\text{edges}} L \cdot \chi(\theta)$ with $\chi(-w) = -\chi(w)$ — i.e. exactly the Φ/colouring method, and exactly Beeson's — is equivalent to the master identity

$$m\big(-c\, z^4 + a\, z^3 + b\, z^2 + a\, z - c\big) = z^3(f - ez)P(z) + (fz - e)Q(z), \qquad P, Q \in \mathbb{Z}[z, z^{-1}],$$

whose $z = \pm 1$ specializations return exactly $M = m(f + e)$ and $-m(f - e)$. But that identity is *solvable for every* m, with the explicit witness $P_0 = mf(z^{-1} - z^{-3})$, $Q_0 = m(ez^2 - fz^3)$. Hence no invariant in this class can force $g \mid M$, for any (e, f, m): the Φ-method — the engine of both Beeson's argument and of our own $2\pi/3$ theorem — provably cannot decide the base-β residue. Closing it requires genuinely non-linear input.

The natural non-linear candidate — the full Conway–Lagarias tiling group, whose abelianization is exactly this Φ class — does not help either. For the base-β tile $(2, 3, 4)$ we computed that group over the finite quotients $\mathbb{F}_p[i]$ ($p \equiv 3 \pmod 4$, 15 a quadratic residue), across $p \in \{43, 59, 67, 71, 103, 127\}$: the class-2 (commutator) layer collapses completely onto the tile relators, so the group reduces to its rank-2 abelianization (M_α, M_β), which is admissible for every candidate. In particular the $m = 1$ boundary word for $N = 11$ is trivial in the group — indistinguishable from the $m = 2, 3, 4$ words that *do* tile — while the controls hold (a triangle not congruent to the tile is non-trivial, and the tileable $N = 44, 99, 176$ words are trivial). So the tiling group is not an obstruction here, extending the collapse we observed for the $(8, 7, 13)$ tile (Remark 79) to the base-β tile itself (`code/verify_tiling_group_basebeta.py`). The non-linear input that closes the branch, if any, is not homological.

Finally, the standard *interior* tool fails here too, and for a reason that is an identity rather than an accident. Beeson's Γ_c argument — the engine of his equilateral no-prime theorem — runs by showing Γ_c is empty: if no relation $jc = ua + vb$ is witnessed with j below the tile count, a Γ_c link must emanate from the boundary, which is impossible since links have in- and out-degree 1 and cannot terminate on ∂ABC. For the base-β tile that hypothesis is false uniformly. Reducing $jc = ua + vb$ modulo f twice gives $f \mid v$, then $f \mid u - we$, and the complete solution set

$$v = fw, \qquad u = we + fs, \qquad j = se + wf \qquad (w, s \in \mathbb{Z}_{\geq 0}),$$

generated by

$$e\, c = f\, a \quad (j = e), \qquad\qquad f\, c = e\, a + f\, b \quad (j = f).$$

The first is the identity $ef^2 = f \cdot ef$. So a relation with $j = e$ always exists, and $e < f \ll N = 3f^2 - e^2$; the Γ_c-emptiness contradiction therefore never triggers, for any (e, f). The classification and the witness are machine-checked in Lean (`lean/GammaC.lean`: `gammac_classification`, `gammac_witness`, `gammac_j_lt_N`; axiom-clean). Closing this branch needs control of a *non-empty* Γ_c, which is exactly the step left open in [4].

The three failures above share one feature, and that feature settles what any future argument may look like. Fix the tile $(2, 3, 4)$, which is the base-β tile at $(e, f) = (1, 2)$, and consider its targets $(8m, 8m, 11m)$ with $N = 11m^2$. These targets are pairwise similar and the tile is the same in each. The case $m = 1$ ($N = 11$) admits no tiling, by exhaustive search; the cases $m = 2$ and $m = 3$ do admit one, by the kernel-checked certificates of Remark 21. Hence, if $\mathcal{C}$ is any criterion for non-tileability whose hypotheses depend only on the similarity class of the tile and that of the target, taken separately, then $\mathcal{C}$ cannot exclude the base-β target at $m=1$: its hypotheses hold verbatim at $m=2$, where a tiling exists, so $\mathcal{C}$ would contradict it. The statement is formalized and axiom-free (`lean/ScaleRigidity.lean`: `no_similarity_invariant_proof`); only the *existence* at $m=2$ is used, never the search result at $m=1$.

Angle arithmetic, the direction group $\langle \eta, -1\rangle$, the Φ class, the angular layer of the Conway–Lagarias group, and any Dehn-type invariant are all criteria of this kind. So the three obstructions recorded above are not three accidents but one, and the same applies to any further invariant assembled from the same data. What survives is exactly the scale-sensitive information carried by the edge-walk conditions, which do separate the cases: for the tile $(2,3,4)$ every admissible walk on an equal side is free of b-edges when $m=1$, whereas at $m=2$ the walk $(P,Q,R)=(3,2,1)$ is admissible and carries two of them (both machine-checked, ibid.). Any proof of the base-β no-go must take such input.

Remark 21 (Beeson's Theorem 14 is *false*, not merely unsound)**.** The exhaustive engine settles the status of $g \mid M$ outright. The isosceles triangle $(24,24,33)$ with base angles β *is* tiled by 99 copies of the tile $(2,3,4)$: this is the base-β family at $(e,f,m)=(1,2,3)$, so $X=f^3m=24$, $Y=em(3f^2-e^2)=33$, $N=m^2(3f^2-e^2)=99$ and $M=(f+e)m=9$. The tiling was found by exact search, re-verified independently (congruence of all 99 tiles, disjointness, containment, exact area) against a separately constructed instance, and is now *machine-verified in Lean 4 with no axioms whatsoever* (`lean/Tiling99.lean`: the kernel checks, by pure computation over $\mathbb{Z}[\sqrt{15}]$ after scaling by 16, that all 99 triangles have squared side multiset exactly $\{1024,2304,4096\}=(16\cdot(2,3,4))^2$, are positively oriented, lie in the closed target $(384,384,528)=16\cdot(24,24,33)$, admit an explicit separating edge-line for each of the 4851 pairs, and have signed areas summing exactly to the target's; `#print axioms` reports the theorem depends on no axioms). Since it is this object that refutes [3, Thm. 14], the refutation now rests on a kernel-checked certificate rather than on a search log. Its boundary base walk is the multiset $\{a^2,b^7,c^2\}$ $(2\cdot2+7\cdot3+2\cdot4=33)$, consistent with $q_{\text{base}}\equiv em\equiv 1 \pmod 2$.

But $g=\gcd(a,c)=\gcd(2,4)=2$ and $M=9$, so $g\nmid M$. Hence

[3, Thm. 14]**'s conclusion** $g \mid M$ **is FALSE**,

not merely unsoundly proved (Remark 13); equivalently $f \mid m$ fails, here $2\nmid 3$. Three consequences. (i) The base-β prime exclusion cannot rest on Theorem 14 *at all*; only a per-value search remains, which is what we do. (ii) Our own base-$(\alpha{+}\beta)$ and base-α replacements (Propositions 30, 12) are unaffected: both were deliberately built to use *no* $g \mid M$. (iii) The values whose exclusion had rested only on $g\mid M$ $(21,39,66,70)$ are demoted (Remarks 70, 13); 21 and 39 are now excluded by exhaustive search instead. Finally $N=99$ is itself a new realization: 99 is not a square, a sum of two squares, nor $2n^2,3n^2,6n^2$, so it is not commensurably realizable.

Proposition 22 (Dissection basics)**.** *Let ABC be cut into N triangles $T_1,\dots,T_N$ with pairwise disjoint interiors and union ABC. Then $\sum_i|T_i|=|ABC|$, and if the T_i are pairwise congruent to a tile T then $|ABC|=N|T|$. Moreover, at a point $v\in ABC$ the tiles whose closure contains v subtend angles summing to 2π if v is interior to ABC, to π if v is interior to a side, and to the corner angle if v is a corner; each summand is a tile angle α,β,γ if v is a vertex of that tile, and π if v lies in the relative interior of one of its edges. Hence the sum has the form $p\alpha+q\beta+r\gamma+s\pi$ with $p,q,r,s\ge 0$ integers.*

Proof. The area identity is machine-checked in `lean/Dissection.lean` (`Dissection.volume_target`, `volume_target_of_congruent`), in `EuclideanSpace` $\mathbb{R}^2$ with Lebesgue measure and axiom-clean: the tiles are convex with positive volume and null frontier (`Convex.addHaar_frontier`), so the union is almost-everywhere disjoint and volumes add. The angle statement is the local structure of a dissection: a small disc about v meets finitely many tiles, each in a single circular sector, and the sectors tile the disc's intersection with ABC. The $s\pi$ term is the contribution of tiles meeting v in an edge interior; it is present exactly when the dissection is not edge-to-edge at v, which does occur here (Remark 18). Mathlib has no theory of planar dissections, so this angle statement is not formalized; it is the geometric input flagged in Section 8. What *is* formalized is the arithmetic consequence drawn from it in Proposition 23 (`vertex_multiplicities` and its three corollaries) and the angle sum of a tile (`cornerAngle_sum`). □

Proposition 23 (Vertex figures). *Let the tile satisfy $3\alpha + 2\beta = \pi$ with $\alpha \notin \mathbb{Q}\pi$, so $\gamma = 2\alpha + \beta$ and $2\gamma = \pi + \alpha$. If the angles at v have multiplicities (p, q, r) for (α, β, γ) together with s straight angles, then $p\alpha + q\beta + r\gamma = t\pi$ with $t = 1$ or $t = 2$, and*

$$t = 1 : \ (p, q, r) \in \{(3, 2, 0), (1, 1, 1)\}; \qquad t = 2 : \ (p, q, r) \in \{(6, 4, 0), (4, 3, 1), (2, 2, 2), (0, 1, 3)\}.$$

A junction interior to a side of ABC has $t = 1$; an interior vertex has $t = 2 - s$, so $t \in \{1, 2\}$. In particular, in every case $r \le 1$ when $t = 1$.

Proof. Write $\beta = (\pi - 3\alpha)/2$ and $\gamma = (\pi + \alpha)/2$; then $p\alpha + q\beta + r\gamma = \frac{1}{2}\big((2p - 3q + r)\alpha + (q + r)\pi\big)$. By Proposition 22 this equals $(t - s')\pi$ for integers, and since α/π is irrational (`tile_alpha_irrational`) both $2p - 3q + r = 0$ and $q + r = 2t$ follow. This is exactly `vertex_multiplicities`, machine-checked. Enumerating the non-negative solutions of $q + r = 2t$, $2p = 3q - r$ gives the two lists (`vertex_pi`, `vertex_beta_corner`, `vertex_apex` in `BaseBetaE1.lean`); that a geometric vertex figure supplies those integer equations is `vertex_pi_multiplicities`, `vertex_beta_corner_multiplicities` and `vertex_apex_multiplicities` in `Dissection.lean`. □

Proposition 24 (Corner figures). *On a base-β target, the vertex figure at each base corner is a* single *tile presenting β, and the figure at the apex is* exactly three *tiles, each presenting α. In particular no corner of ABC carries a γ. The two edges of the base-corner tile at that corner are a and c; the edges of the apex tiles at the apex are b and c.*

Proof. By Proposition 23, an angle relation $p\alpha + q\beta + r\gamma = t$ at a corner splits into $2p - 3q + r = \sigma$ and $q + r = \tau$ for integers determined by t (the splitting step is machine-checked: `lean/Dissection.lean`, `vertex_multiplicities` and `vertex_pi_multiplicities`, from $3\alpha + 2\beta = \pi$, $\gamma = 2\alpha + \beta$ and irrationality of α/π via `angle_indep`). A base corner has $t = \beta$, giving $\sigma = -3$, $\tau = 1$; the case $(q, r) = (0, 1)$ would need $2p = -4$, so $(p, q, r) = (0, 1, 0)$. The apex has $t = 3\alpha$, giving $\sigma = 6$, $\tau = 0$, hence $q = r = 0$ and $p = 3$. The edge assertions are the incidence rule: a joins the β, γ corners, b joins α, γ, c joins α, β. Machine-checked (`corner_beta_unique`, `corner_apex_unique` in `lean/BaseBetaCorners.lean`). □

Proposition 25 (γ-trap). *Every side of a base-β target carries at least one c-edge.*

Proof. A junction interior to a side is a π-vertex, so by Proposition 23 its multiplicity vector is $(3, 2, 0)$ or $(1, 1, 1)$: it carries *at most one* γ (`pi_vertex_gamma_le_one`).

Let a side of ABC be partitioned by the tile edges $E_1, \dots, E_n$ lying on it, with junctions $J_0, \dots, J_n$, where J_0 and J_n are corners of ABC. (The partition is exact: the tiles cover ABC with disjoint interiors, so each point of the side lies in exactly one tile edge. This is a statement about the boundary and is unaffected by non-edge-to-edge incidence in the interior.)

Suppose the side carries no c-edge, so every E_i is an a- or a b-edge. Both are incident to γ at exactly one endpoint, so each E_i places a γ at J_{i-1} or at J_i, and by the previous paragraph no junction receives two. By Proposition 24 neither J_0 nor J_n carries a γ at all. Hence E_1 must place its γ at J_1; that junction is then full, so E_2 must place at J_2; inductively E_i places at J_i, and finally E_n places at J_n — a corner. Contradiction. □

Remark 26. The hypothesis that the corner angles are $\beta, \beta, 3\alpha$ is essential, and this is why [3, Lem. 6] is false as quoted (Remark 9): on the tile-similar target a corner *is* a γ, the chain terminates harmlessly, and the counterexample there has two sides with no c-edge. On a base-β target no corner is a γ, and the argument runs.

Proposition 27 (b is unsplittable)**.** *b is not a sum of two or more tile edges: if $n_a\, a+n_b\, b+n_c\, c = b$ with $n_a, n_b, n_c \geq 0$, then $(n_a, n_b, n_c) = (0, 1, 0)$.*

Proof. If $n_b = 0$ the remaining terms are divisible by f, so $f \mid b$; with $b + e^2 = f^2$ this gives $f \mid e^2$, and $\gcd(e, f) = 1$ forces $f = 1$, contradicting $1 \leq e < f$. If $n_b \geq 2$ then $n_b\, b > b$ already, as $b > 0$. So $n_b = 1$, the remaining terms sum to zero, and $a, c > 0$ give $n_a = n_c = 0$. Machine-checked (`b_unsplittable`). □

Proposition 28 (Corner parallelogram)**.** *Let T_A be the tile at a base corner A. Its b-edge is a chord $[P, Q]$ with both endpoints on ∂ABC, matched by exactly one tile T_3, which presents α where T_A presents γ and conversely; so $T_A \cup T_3$ is a parallelogram with sides a and c. Consequently the first* two *edges of each of the two sides meeting at A lie in $\{a, c\}$.*

Proof. By Proposition 24, T_A is the unique tile at A and its two edges there are a and c, one along each side of ABC; so its other two vertices P (the γ corner) and Q (the α corner) lie one on each of those two sides. Its third edge is the chord $[P, Q]$.

The tiles on the far side of $[P, Q]$ have edges covering it. No such edge can straddle: a straddling edge would extend beyond P or beyond Q along the line PQ, and both P and Q lie on ∂ABC, so the extension leaves ABC. Hence the far-side edges partition $[P, Q]$ exactly, and by Proposition 27 the partition is trivial: a single tile T_3 with b-edge $[P, Q]$.

T_A presents γ at P and α at Q. Both P and Q are interior to sides of ABC, hence π-vertices, which carry at most one γ; since T_3's b-edge is incident to α and γ, it must present α at P and γ at Q. Two congruent triangles glued along b with α, γ interchanged form a parallelogram with sides a, c.

For the last assertion, take the side on which Q lies and let E_1 be its first edge (an edge of T_A). At Q the angles are T_A's α, T_3's γ, and, since $\alpha + \gamma = \pi/2 + 3\theta$, a remaining $\beta = \pi/2 - 3\theta$ carried by one further tile. That tile owns E_2 and presents β, whose incident edges are a and c. The same argument at P handles the other side. □

TABLE 1. Base-β members with $N \leq 110$: every one is settled by exhaustive search (corner-anchored exact search over $\mathbb{Q}(\sqrt{D})$, C++ engine, node-for-node validated against the independent Python engine). "thin" means $f > 2e$.

N	(e, f)	tile	regime	search verdict	(nodes)
11	$(1, 2)$	$(2, 3, 4)$	thick	no tiling	135
23	$(2, 3)$	$(6, 5, 9)$	thick	no tiling	19 677
26	$(1, 3)$	$(3, 8, 9)$	thin	no tiling	2 025
39	$(3, 4)$	$(12, 7, 16)$	thick	no tiling	825 724
47	$(1, 4)$	$(4, 15, 16)$	thin	no tiling	12 440
59	$(4, 5)$	$(20, 9, 25)$	thick	no tiling	1 838 175
66	$(3, 5)$	$(15, 16, 25)$	thick	no tiling	7 232 464
71	$(2, 5)$	$(10, 21, 25)$	thin	no tiling	553 417
74	$(1, 5)$	$(5, 24, 25)$	thin	no tiling	132 519
83	$(5, 6)$	$(30, 11, 36)$	thick		searching
107	$(1, 6)$	$(6, 35, 36)$	thin	no tiling	1 251 382

Remark 29 (The colouring theorem is citation-free)**.** The same identity has a positive consequence: it *proves* the sound half of [3, Thm. 14] rather than citing it. Because α/π is irrational (Remark 20's η, via Niven — `tile_alpha_irrational`), η is not a root of unity: $n(\frac{\alpha}{2}+\frac{\pi}{2}) = k\pi$ forces $n = 0$ (`direction_free`, `colouring_well_defined`, both axiom-clean). Hence $\langle \eta, -1\rangle \cong \mathbb{Z}\times\mathbb{Z}/2$ and the sign character $\chi_2(\pm\eta^n) = \pm(-1)^n$ is *well defined* — which is exactly what makes the colouring number $M = \sum_t \chi_2(d_t)$ a well-defined integer, and the $z = -1$ specialization of the master identity then returns $M = m(f+e)$ together with the size formulas $X = f^3M/(f+e)$, $Y = eM(3f^2-e^2)/(f+e)$. Consequently the $3\alpha+2\beta$ tiling equations — and with them the necessary sides of the triquadratic and four-component branches (Remark 11) — no longer rest on a preprint's colouring theorem: the direction-group foundation is machine-checked, and the cancellation step is the same Φ-style argument proved in Section 4.

Proposition 30 (Base-($\alpha+\beta$) prime exclusion, correcting [3, Thm. 18])**.** *No isosceles triangle with base angles $\alpha+\beta$, where $3\alpha+2\beta = \pi$ and $\alpha \notin \mathbb{Q}\pi$, is N-tiled for N prime.*

Proof. Fix a putative tiling with coloring number M. On this target [3, Thm. 17] gives $N/M^2 = (f+e)/(f-e) > 1$, hence $0 < M^2 < N$; we use that rather than [3, Cor. 3], which is quoted in the literature without scope and cannot hold in the stated generality [1, Rem. 2.1]. Put $d = \gcd(N-M^2, N+M^2)$. As $d \mid 2N$, $d \mid 2M^2$, and $\gcd(N,M^2) = 1$ (a prime $N > M^2$ cannot divide M^2), $d \mid 2$. Writing $g = (N+M^2)/d$ and $\hat{a} = (N-M^2)/d$, the primitive tile is $(a,b,c) = (g\hat{a},\ g^2-\hat{a}^2,\ g^2)$ with $0 < \hat{a} < g$ ([3, Thm. 17]; we do not use [3, Lem. 8], whose squarefree half this paper shows to be false — only the parametrization is needed, and it also follows from the $3\alpha+2\beta$ relation directly), and the base AC has integer length $Y = M\hat{a}(g+\hat{a})$ ([3, Lem. 42]). A tiling forces the base to carry at least *one* c-edge — this is the γ-trap, Proposition 25, proved here from the vertex figures alone and machine-checked (`gamma_injection`, `c_edge_exists`) — and a number of b-edges $\equiv -M \pmod g$, hence at least $g - M > 0$ ([3, Lem. 45(iii)]); so $Y \geq (g-M)b + c$. But $d \mid 2$ gives $\hat{a} + M \leq g$, and the identity

$$(g-M)(g^2-\hat{a}^2) - M\hat{a}(g+\hat{a}) = g\,(g+\hat{a})\,(g-\hat{a}-M)\ \geq\ 0$$

gives $Y \leq (g-M)b < (g-M)b + 2c$, a contradiction. (E.g. $N = 79$, $M = 1$: $d = 2$, $g = 40$, $\hat{a} = 39$, $b = 79$, $c = 1600$, so $Y = 3081$ while $(g-M)b + 2c = 6281$.) The two arithmetic facts used — $d \mid 2$ and the identity together with $\hat{a} + M \leq g$ — are machine-checked in Lean (`lean/BaseAlphaBetaPrime.lean`, axiom-clean); only the covering bounds are cited. ∎

The tile. Throughout, T has angles (α,β,γ) opposite sides (a,b,c), with $\gamma = 2\pi/3$, hence $\alpha+\beta = \pi/3$; we treat the incommensurable case, α an irrational multiple of π. Once the target is known isosceles (Section 3), T has commensurable sides by [4, Thm. 12.5] (refereed); we scale to

$$(a,b,c) \in \mathbb{Z}^3, \qquad \gcd(a,b,c) = 1, \qquad c^2 = a^2 + ab + b^2 \tag{1}$$

(the law of cosines at $\gamma = 2\pi/3$). We call such a triple a *primitive* $120°$*-triple.* If a prime p divides two of a,b,c it divides the third (by (1)), so (1) gives $\gcd(a,b) = \gcd(a,c) = \gcd(b,c) = 1$. One computes

$$\sin\alpha = \tfrac{a\sqrt{3}}{2c}, \quad \cos\alpha = \tfrac{2b+a}{2c}, \qquad \sin\beta = \tfrac{b\sqrt{3}}{2c}, \quad \cos\beta = \tfrac{2a+b}{2c}. \tag{2}$$

The direction grid. Set $G = \{\, j\frac{\pi}{3} + k\alpha : j,k \in \mathbb{Z}\,\} \pmod{2\pi}$. We claim every edge of every tile in a tiling points in a direction lying in G, for a single fixed reference. Fix the reference along an edge of one tile; its three edge-directions are then in G, since the tile angles $\alpha, \beta = \frac{\pi}{3} - \alpha, \gamma = \frac{2\pi}{3}$ all lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$. Now argue by adjacency. The tiling fills a connected region, so any tile t' shares at least a boundary point P with the union of the tiles already placed; at P the surrounding corner angles each belong to $\{\alpha,\beta,\gamma\}$, and a directed edge of t' at P differs from a known grid direction at P by a sum of such angles, hence lies in G; the remaining edges of

t' then differ from it by $\mathbb{Z}$-combinations of $\alpha, \frac{\pi}{3}$, so all of t' is in G. This uses only the angles meeting at a point and so applies verbatim at non-edge-to-edge incidences, where P lies in the relative interior of an edge (the angles there sum to π rather than 2π). Since α/π is irrational, $\frac{\pi}{3}$ and α are $\mathbb{Q}$-linearly independent, so the pair (j,k) is determined by the direction (and $(-1)^j$ by the direction alone, as $2\pi = 6 \cdot \frac{\pi}{3}$).

Vertex types and shapes of ABC. A corner of ABC is filled by tile-corners: if $P, Q, R \geq 0$ (not all 0) of them are α-, β-, γ-corners, the corner measures $P\alpha + Q\beta + R\gamma = (P-Q)\alpha + (Q+2R)\frac{\pi}{3}$. Write $(m,k) := (P-Q,\ Q+2R)$. Since $Q \equiv k \pmod 2$, $0 \leq Q \leq k$, and $P = m+Q \geq 0$, the pair (m,k) is *realizable* exactly when

$$k \geq 1 \text{ and } m \geq -k, \qquad \text{or} \qquad k = 0 \text{ and } m \geq 1. \tag{3}$$

The three corners of ABC have angles $m_i\alpha + k_i\frac{\pi}{3}$ summing to $\pi = 3 \cdot \frac{\pi}{3}$; as α is irrational this forces

$$\textstyle\sum_i m_i = 0, \qquad \sum_i k_i = 3. \tag{4}$$

By (3) each $m_i + k_i \geq 0$, and by (4) $\sum_i (m_i + k_i) = 3$; hence each $m_i + k_i \in \{0,1,2,3\}$ and $m_i \in [-k_i,\ 3-k_i] \subseteq [-3,3]$. The enumeration is therefore finite and *independent of* α. Checking the finitely many triples $\{(m_i,k_i)\}$ satisfying (3), (4) and $0 < m_i\alpha + k_i\frac{\pi}{3} < \pi$ (done in exact arithmetic in Section 8) leaves exactly eleven shapes of ABC: the equilateral triangle; two isosceles triangles, with base angles α (apex $\alpha + 3\beta$) and β (apex $\beta + 3\alpha$); and eight scalene triangles, namely

$$F_1 = (\alpha, \alpha+\beta, \alpha+2\beta), \quad F_2 = (2\alpha, 2\beta, \alpha+\beta), \quad F_3 = (\alpha, 2\alpha, 3\beta), \quad F_4 = (\alpha, 2\beta, 2\alpha+\beta),$$

their reflections under $\alpha \leftrightarrow \beta$ (with F_2 self-paired), and the tile-similar triangle $(\alpha, \beta, 2\pi/3)$. Since α/π is irrational, distinct pairs (m,k) give distinct angles, so the labels *equilateral*, *isosceles*, *scalene* attached to these types are unambiguous: no accidental coincidence of angles can occur, as any such coincidence would force $\alpha \in \mathbb{Q}\pi$. These types coincide with the six sporadic non-equilateral $2\pi/3$ similarity types of Zhang [9].

The area identity. If ABC is similar to a fixed shape whose primitive integer side-proportion vector is D (gcd of its entries 1), and ABC is N-tiled, then each side of ABC is a disjoint union of whole tile-edges (a tile meeting a straight portion of the boundary has a whole side on it), hence an integer; so the sides of ABC are $k \cdot D$ for an integer $k \geq 1$, and

$$N = \frac{\mathrm{Area}(ABC)}{\mathrm{Area}(T)} = k^2 N_0, \qquad N_0 := \frac{\mathrm{Area}(D)}{\mathrm{Area}(T)}, \qquad \mathrm{Area}(T) = \tfrac{\sqrt{3}}{4}ab. \tag{5}$$

This is a necessary condition on N.

3. Reduction to the isosceles case

The reduction below takes the tile to be integer-sided. That input is a citation ([4, Thm. 12.5] on the isosceles targets, [8] in general). The following proposition recovers it from the invariant alone, with no citation, on any target where the invariant takes integer values — which is how it is used in Sections 4 onwards. We state it first because it is logically prior to everything that follows.

Proposition 31 (Rationality from integrality of the invariant)**.** *Let T have sides a, b, c with $c^2 = a^2 + ab + b^2$ (equivalently, the angle opposite c is $2\pi/3$), let $P \neq 0$ be a real scale, and suppose the three quantities*

$$N = P^2\, b(a+2b), \qquad M_\alpha = P(c-a-b), \qquad M_\beta = P(c+a+b)$$

are all integers. If $b \neq 0$ and $a+b \neq 0$, then $a/b \in \mathbb{Q}$ and $c/b \in \mathbb{Q}$: the tile is rational up to scale.

Proof. Two identities. First, $M_\alpha M_\beta = P^2\big(c^2 - (a+b)^2\big) = -P^2ab$ by the tile relation, whence

$$N + M_\alpha M_\beta = P^2 b(a+2b) - P^2ab = 2P^2b^2 \neq 0,$$

and $a\,(N + M_\alpha M_\beta) = 2P^2ab^2 = b\,(-2M_\alpha M_\beta)$, so

$$\frac{a}{b} = \frac{-2M_\alpha M_\beta}{N + M_\alpha M_\beta} \in \mathbb{Q}.$$

Second, $M_\beta - M_\alpha = 2P(a+b) \neq 0$ and $M_\alpha + M_\beta = 2Pc$, so $c\,(M_\beta - M_\alpha) = (a+b)(M_\alpha + M_\beta)$ and

$$\frac{c}{a+b} = \frac{M_\alpha + M_\beta}{M_\beta - M_\alpha} \in \mathbb{Q}.$$

Multiplying by $(a+b)/b = a/b+1 \in \mathbb{Q}$ gives $c/b \in \mathbb{Q}$. Machine-checked in `lean/Rationality.lean` (`tile_rational`), axiom-clean. □

Remark 32. The proposition converts the rationality input from a hypothesis about the tile into a hypothesis about the invariant, and the latter is what the method produces natively: wherever the boundary functional of Section 4 is defined and integral, rationality is automatic. It does not by itself discharge the citation on the *general* target, because establishing integrality of all three quantities there is exactly the content of the cited theorem. What it does discharge is the appearance of a rationality assumption *inside* the invariant argument, where the three integralities are already in hand; and it shows the two facts are equivalent rather than independent, which is not how the literature presents them.

Proposition 33. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. If T N-tiles a triangle and N is prime, then ABC is isosceles and not equilateral.*

Proof. By Section 2, the corner-type triple of ABC is the equilateral one, the tile-similar one, one of $F_1, \dots, F_4$ or their reflections, or one of the two isosceles ones. The equilateral case gives N not prime [6, Thm. 6]; the tile-similar case gives $N = n^2$ [2, 14], not prime. For the remaining non-isosceles shapes we compute N_0 from (5); in each case $N = k^2N_0$ is composite.

F_2 and F_4. The side-proportion vector of F_2 is $\big(a(a+2b),\, b(2a+b),\, c^2\big)$ with the $\pi/3$ vertex between the first two sides, giving $N_0 = (a+2b)(2a+b)$. For F_4 one checks the side-vector $\big(ac,\, b(2a+b),\, (a+b)c\big)$ has gcd 1 (any common prime g divides c and then $g^2 \mid 3b^2$, forcing $g = 1$), giving $N_0 = (a+b)(2a+b)$. Both are products of two integer factors each at least 3, hence composite, so $N = k^2N_0$ is composite.

F_3. The law of sines gives the side-proportion vector $\big(ac^2,\, a(a+2b)c,\, 3ab(a+b)\big)$, whose entries share the factor a; dividing it out leaves $\big(c^2,\, (a+2b)c,\, 3b(a+b)\big)$, and $N_0 = 3(a+b)(a+2b)$. This is divisible by 3, hence composite, so $N \equiv 0 \pmod 3$ is composite.

F_1. The side-vector is $(a, c, a+b)$ with the $\pi/3$ vertex between the sides a and $a+b$, so $N_0 = (a+b)/b$ and $N = k^2(a+b)/b$. As $\gcd(b, a+b) = 1$, integrality forces $b \mid k^2$, so $N = t(a+b)$ with $t = k^2/b \in \mathbb{Z}_{\geq 1}$. It remains to show $a+b$ is composite. Suppose $a+b = p$ is prime. Since $(a+b)^2 - c^2 = ab$ by (1), we have $ab = p^2 - c^2$, so a, b are the roots of $x^2 - px + (p^2 - c^2)$, whose discriminant $4c^2 - 3p^2$ must be a perfect square d^2; then $3p^2 = (2c-d)(2c+d)$. With $2c - d < 2c + d$ the factorizations of $3p^2$ into two positive factors are $(1, 3p^2)$, $(3, p^2)$, $(p, 3p)$, and in each the smaller root $\frac{p-d}{2}$ is not a positive integer: in the first, $d = (3p^2-1)/2 > p$, so the root is negative; in the second, $d = (p^2-3)/2$ and the root $\frac{p-d}{2} = -\frac{(p-3)(p+1)}{4}$ is negative for $p > 3$; in the third, $d = p$ gives $c = p$ and $ab = p^2 - c^2 = 0$. (The case $p \leq 3$ is impossible

since $a+b \geq 8$ for a primitive 120°-triple.) Hence $a+b \geq 8$ is composite, and $N = t(a+b)$ is composite.

The reflected shapes are handled by $\alpha \leftrightarrow \beta$ (which swaps $a \leftrightarrow b$ and leaves each N_0 above of the same form). So a prime N forces the corner-type triple of ABC to be one of the two isosceles types; those have angles $(\varphi, \varphi, \pi - 2\varphi)$ with $\varphi \in \{\alpha, \beta\}$, and they are never equilateral ($\varphi = \frac{\pi}{3}$ would force $\beta = 0$ or $\alpha = 0$). Hence ABC is isosceles and not equilateral. □

4. The signed-direction invariant

Fix the reference so the direction grid is $G = \{\, j\frac{\pi}{3} + k\alpha \,\}$. Because $\beta = \frac{\pi}{3} - \alpha$, every grid direction also has the form $(j+k)\frac{\pi}{3} - k\beta$, so we may read off the coefficient of $\frac{\pi}{3}$ in two coordinate systems. Define, for a direction $\theta \in G$,

$$f_\alpha(\theta) = (-1)^j \quad (\theta = j\tfrac{\pi}{3} + k\alpha), \qquad f_\beta(\theta) = (-1)^{j'} \quad (\theta = j'\tfrac{\pi}{3} + k'\beta).$$

Both are well defined (Section 2), and both satisfy

$$f(\theta + \pi) = -f(\theta), \tag{6}$$

since adding $\pi = 3 \cdot \frac{\pi}{3}$ increases the $\frac{\pi}{3}$-coefficient by 3. They are genuinely different functions: $f_\beta = (-1)^k f_\alpha$.

For $f \in \{f_\alpha, f_\beta\}$ and a directed edge of length L and direction θ, assign the weight $L\,f(\theta)$. For a tile t with boundary traversed counterclockwise, put $C_f(t) = \sum_{\text{edges}} L\,f(\theta)$, and for the boundary of ABC put $\Phi_f(\partial ABC) = \sum_{\text{sides}} L\,f(\theta)$ (counterclockwise).

Lemma 34 (Cancellation)**.** *For each $f \in \{f_\alpha, f_\beta\}$, $\displaystyle\sum_{\textit{tiles}} C_f(t) = \Phi_f(\partial ABC)$.*

Proof. For a grid direction θ let $\Lambda(\theta)$ be the total length of all directed tile-edges of direction θ, summed over every tile (counterclockwise) and every edge; then, grouping the double sum $\sum_t C_f(t)$ by direction,

$$\sum_{\text{tiles}} C_f(t) = \sum_\theta \Lambda(\theta)\, f(\theta). \tag{7}$$

Take the *common refinement*: cut every tile-edge at every point where any other tile-edge meets it, so that the edges of all tiles are subdivided into finitely many *minimal pieces*, each of which is either interior to ABC or contained in ∂ABC, and no two of which overlap partially. Refining the sums in this way changes nothing, since the weight is linear in length. (Working with minimal pieces rather than with maximal straight segments is what makes the next step airtight: a minimal piece is covered by whole tile-edge pieces on each side, with no overhang.) Let S be a minimal piece. If S lies in the interior of ABC, a neighbourhood of a generic point of S is a disc split by S into two half-discs, each tiled; the tiles on one side meet S in directed edges of total length $|S|$ pointing one way, say θ, and those on the other side in directed edges of total length $|S|$ pointing $\theta + \pi$ (the two subdivisions of the ambient segment may differ — that is exactly a non-edge-to-edge incidence — but each side covers the minimal piece S exactly once, because the tiles cover ABC with disjoint interiors and S is not partially overlapped by construction). If instead $S \subseteq \partial ABC$, only the single tile inside ABC meets it, contributing length $|S|$ in the counterclockwise direction of ∂ABC. Writing $\Lambda = \Lambda_{\text{int}} + \Lambda_{\text{bd}}$ accordingly, interior segments give $\Lambda_{\text{int}}(\theta) = \Lambda_{\text{int}}(\theta + \pi)$ for every θ, while $\Lambda_{\text{bd}}(\theta)$ is the counterclockwise boundary length in direction θ. Substituting into (7) and pairing θ with $\theta + \pi$,

$$\sum_{\{\theta, \theta+\pi\}} \Lambda_{\text{int}}(\theta)\big(f(\theta) + f(\theta + \pi)\big) = 0$$

by (6), so $\sum_{\text{tiles}} C_f(t) = \sum_\theta \Lambda_{\text{bd}}(\theta) f(\theta) = \Phi_f(\partial ABC)$. □

Lemma 35 (Tile value)**.** *For every placement of T, $C_{f_\alpha}(t) = \pm(c + a - b)$ and $C_{f_\beta}(t) = \pm(c + b - a)$.*

Proof. Take a positively oriented placement and label the vertices V_1, V_2, V_3 in counterclockwise order with interior angles α, β, γ respectively; let e_i be the directed edge from V_i to V_{i+1} (indices mod 3). Then e_1, e_2, e_3 are opposite V_3, V_1, V_2, so they have lengths c, a, b. Traversing counterclockwise turns left at V_{i+1} by the exterior angle $\pi -$(angle at V_{i+1}), so $\mathrm{dir}(e_2) = \mathrm{dir}(e_1) + (\pi - \beta)$ and $\mathrm{dir}(e_3) = \mathrm{dir}(e_2) + (\pi - \gamma)$. Now π adds 3 to the $\frac{\pi}{3}$-coefficient j, while α adds 0, $\beta = \frac{\pi}{3} - \alpha$ adds 1, and $\gamma = \frac{2\pi}{3}$ adds 2; so $\pi - \beta$ adds $3 - 1 = 2$ and $\pi - \gamma$ adds $3 - 2 = 1$. Writing j_0 for the $\frac{\pi}{3}$-coefficient of $\mathrm{dir}(e_1)$, the three coefficients are j_0, $j_0 + 2$, $j_0 + 3$, so $f_\alpha = (-1)^{j_0}$ on e_1, e_2 and $-(-1)^{j_0}$ on e_3. As e_1, e_2, e_3 carry lengths c, a, b,

$$C_{f_\alpha}(t) = (-1)^{j_0}\big(c + a - b\big).$$

Any placement is obtained from this one by a grid rotation $m\alpha + n\frac{\pi}{3}$, which shifts every j by n and multiplies C_{f_α} by $(-1)^n$, possibly followed by a reflection, which reverses the traversal and negates C_{f_α}; hence $C_{f_\alpha}(t) = \pm(c+a-b)$ in every case. Repeating in the β-frame (where $\gamma = \frac{2\pi}{3}$ adds 0 to j' and α adds 1) makes the a-edge the odd one out, giving $C_{f_\beta}(t) = \pm(c + b - a)$. $\square$

Corollary 36 (Integrality and parity)**.** *Each of $M_\alpha = \dfrac{\Phi_{f_\alpha}(\partial ABC)}{c + a - b}$ and $M_\beta = \dfrac{\Phi_{f_\beta}(\partial ABC)}{c + b - a}$ is an integer, and moreover*

$$M_\alpha \equiv M_\beta \equiv N \pmod 2.$$

They are separate necessary conditions: each follows from Lemma 34 for its own f, so a single failure forbids a tiling.

Proof. By Lemmas 34 and 35, $\Phi_{f_\alpha}(\partial ABC) = \sum_t C_{f_\alpha}(t) = (c + a - b)\sum_t \varepsilon_t$ with $\varepsilon_t \in \{\pm 1\}$, so $M_\alpha = \sum_t \varepsilon_t$ is a sum of N signs, whence $M_\alpha \equiv N \pmod 2$; likewise for f_β. $\square$

The product of the two invariants. The two counts are usually treated as independent conditions. They are not: on each of the eleven shapes their *product* is a fixed rational multiple of the tile count.

Proposition 37 (The invariant product)**.** *Let T be a $2\pi/3$ tile with α/π irrational and let ABC be one of the eleven shapes of Section 2, N-tiled by T. Then, up to the global sign fixed by the choice of reference direction,*

$$M_\alpha M_\beta = \kappa\, N, \qquad \kappa = \begin{cases} 3 & \textit{equilateral}, \\ 1 & \textit{tile-similar},\ F_1,\ F_1', \\ 3ab/\big((a+2b)(2a+b)\big) & F_2, \\ -a/(a+2b) & \textit{iso base-}\alpha,\ F_3,\ F_4', \\ -b/(2a+b) & \textit{iso base-}\beta,\ F_4,\ F_3'. \end{cases}$$

Proof. Write a corner angle of ABC as $m\alpha + k\frac{\pi}{3}$ (Section 2). Traversing ∂ABC counterclockwise, the direction of one side exceeds the previous one by the exterior angle $\pi - \theta$, so in the coordinates $\theta = j\frac{\pi}{3} + k'\alpha$ the pair (j, k') advances by $(3 - k, -m)$ at each corner; the three corner conditions $\sum m_i = 0$, $\sum k_i = 3$ of (4) are exactly the statement that the walk closes up, $(j, k') \mapsto (j + 6, k')$. This determines $f_\alpha = (-1)^j$ and $f_\beta = (-1)^{j+k'}$ on each side, the law of sines gives the side lengths in terms of (a, b, c), and $\Phi_{f_\alpha}\Phi_{f_\beta}/(XY)$ with $XY = 3ab$ is then a rational function of a, b which reduces modulo $c^2 = a^2 + ab + b^2$ to κ times the area ratio. The computation is exact and is carried out for all eleven shapes in `code/verify_product.py`; the two entries used below are also machine-checked in Lean (`lean/InvariantProduct.lean`, axiom-clean). $\square$

Corollary 38 (Tile-similar target, citation-free)**.** *If ABC is similar to T, then $N = M_\alpha^2$. In particular N is a perfect square and is never prime.*

Proof. Here ABC is T scaled by some $\mu > 0$, and its sides are unions of tile edges, so its three directions lie in the grid; Lemma 35 applied to ABC itself gives $\Phi_{f_\alpha}(\partial ABC) = \pm\mu(c+a-b)$, i.e. $M_\alpha = \pm\mu$, while the area identity gives $N = \mu^2$. Machine-checked (`tile_similar_not_prime`). ∎

This re-proves the tile-similar row of Proposition 33 from the invariant alone, in place of [14, 2]. The F_1 row can likewise be made citation-free, and this is worth isolating because it removes a use of the rationality input.

Proposition 39 (F_1 excluded for prime N, without a rationality hypothesis)**.** *Let $N > 3$ be prime. No F_1 target, and no reflected F_1' target, is N-tiled by a $2\pi/3$ tile with α/π irrational. The proof does not assume the tile rational.*

Proof. Label the corners V_1, V_2, V_3 with angles α, $\frac{\pi}{3}$, $\frac{\pi}{3} + \beta$, so that $|V_2V_3|, |V_3V_1|, |V_1V_2|$ are μa, μc, $\mu(a+b)$ for a scale $\mu > 0$. By Proposition 37, $M_\alpha M_\beta = N$; both are nonzero integers, so a prime N forces $\{|M_\alpha|, |M_\beta|\} = \{1, N\}$. Clearing the denominators X, Y against each other using $XY = 3ab$ gives the closed forms

$$M_\alpha\, b = \mu\,(c-a), \qquad M_\beta\, b = \mu\,(c+a)$$

(`F1_Ma`, `F1_Mb`), so $|M_\beta/M_\alpha| = (c+a)/(c-a) > 1$; hence $|M_\alpha| = 1$, $|M_\beta| = N$ and

$$\frac{c+a}{c-a} = N, \qquad \text{that is} \qquad \frac{a}{c} = \frac{N-1}{N+1}$$

(`F1_ratio`, `F1_pin`). Normalize $c = 1$. Then $a = (N-1)/(N+1) \in \mathbb{Q}$ and b is the positive root of $b^2 + ab + a^2 - 1 = 0$, namely $b = \big(\sqrt{N^2+14N+1} - (N-1)\big)/\big(2(N+1)\big)$. So $b \in \mathbb{Q}$ would force $N^2 + 14N + 1 = (N+7)^2 - 48$ to be a perfect square, which holds only for $N \in \{0, 1, 6\}$ (`disc_not_square`). Hence for prime $N > 3$ the tile is *irrational*: $a, c \in \mathbb{Q}$ but $b \notin \mathbb{Q}$.

Under the rationality input that is already the contradiction, and it is a shorter route to the F_1 row than the one in Proposition 33. Without it, continue. From $|M_\alpha| = 1$ we get $\mu = b/(c-a) = b(N+1)/2$, so the three sides are

$$\mu a = \tfrac{N-1}{2}\, b, \qquad \mu c = \tfrac{N+1}{2}\, b, \qquad \mu(a+b) = \tfrac{2N}{N+1},$$

the last rational because $b(a+b) = 1 - a^2$. Each side is an exact union of whole tile edges, say $n_a a + n_b b + n_c c$ with $n_a, n_b, n_c \geq 0$; as $1, a \in \mathbb{Q}$ while $b \notin \mathbb{Q}$, matching rational and irrational parts is forced. The side μa has $n_a = n_c = 0$ and $n_b = \frac{N-1}{2}$, the side μc has $n_b = \frac{N+1}{2}$, and the side $\mu(a+b)$ has $n_b = 0$ with $n_a(N-1) + n_c(N+1) = 2N$, whose only nonnegative solution for $N \geq 5$ is $n_a = n_c = 1$.

So ∂ABC carries exactly N b-edges. Each tile has exactly one b-edge, and distinct boundary edges lie in distinct tiles, so *every* tile has its b-edge on the boundary. Now the corners. At V_1 the relation $P\alpha + Q\beta + R\gamma = \alpha$ forces $(P, Q, R) = (1, 0, 0)$: a single tile, presenting α, whose two edges there are b and c. Its b-edge must lie on the b-only side V_1V_3, so its c-edge is the first edge of V_1V_2; the second and last edge of V_1V_2 is therefore the a-edge, incident to V_2. Let T' be the tile owning it. The b-edge of T' lies on V_2V_3 or on V_3V_1. Either way T' has two vertices on the line V_1V_2 and two on a second line meeting it in a corner of ABC; having only three vertices, T' has one at that corner, incident to both its a- and its b-edge, hence presenting the angle between them, $\gamma = \frac{2\pi}{3}$. But the corner angles there are $\alpha < \frac{\pi}{3}$ and $\frac{\pi}{3}$, a contradiction. The reflected shape is the mirror image under $a \leftrightarrow b$. ∎

Remark 40 (What this does and does not remove)**.** Proposition 37 is a statement about the $2\pi/3$ branch only. It says nothing about the base-β branch of $3\alpha + 2\beta = \pi$, which is a different branch with a different tile and is where the one genuine gap of Theorem 1 lives (Remark 13). What it does buy is a narrowing of the rationality dependence: Corollary 38 and Proposition 39 discharge three of the eleven shapes — tile-similar, F_1, F_1' — with no appeal to [8] or [4, Thm. 12.5], whereas the remaining eight still use an integer-sided tile. The obstruction to doing the same for F_2, F_3, F_4 is visible in the table: there $\kappa \neq \pm 1$, so the product pins only the single ratio a/b, and the law of cosines then leaves c/b a quadratic irrationality that no further invariant condition controls. Establishing rationality on a general target remains exactly the content of the cited theorem.

5. The isosceles obstruction

Lemma 41 (Non-integrality)**.** *Let a, k be positive integers with $\gcd(a, k) = 1$, put $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then $k \nmid (a + b - c)$.*

Proof. From $c^2 = a^2 + ab + b^2$ we get $c > a$ and, since $c^2 < (a + b)^2$, also $c < a + b$. Suppose $k \mid (a + b - c)$. As $b = k^2 \equiv 0 \pmod{k}$, this gives $c \equiv a \pmod{k}$; with $0 < c - a < b = k^2$ we may write $c = a + kt$, $1 \le t \le k - 1$. Substituting into $c^2 = a^2 + ak^2 + k^4$ and dividing by k,

$$2at + kt^2 = ak + k^3, \qquad \text{that is} \qquad a(2t - k) = k(k - t)(k + t).$$

Then $k \mid a(2t - k)$, and $\gcd(a, k) = 1$ gives $k \mid (2t - k)$, hence $k \mid 2t$. But $2 \le 2t \le 2k - 2$ contains no multiple of k other than k itself, so $2t = k$. The left side is then $a(2t - k) = 0$, while the right side is $k \cdot \frac{k}{2} \cdot \frac{3k}{2} = \frac{3k^3}{4} > 0$, a contradiction. Hence $k \nmid (a + b - c)$. ☐

Theorem 42. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. Then no prime number of copies of T tiles an isosceles, non-equilateral triangle.*

Remark 43. The irrationality hypothesis cannot be dropped: the commensurable tile $(\frac{\pi}{6}, \frac{\pi}{6}, \frac{2\pi}{3})$ tiles an equilateral (hence isosceles) triangle with $N = 3$, by the fan from the centre; that tile lives in the commensurable branch of the classification, where $N = 3n^2$ is allowed. The equilateral target for an *incommensurable* $2\pi/3$ tile is excluded for primes by Beeson [6], and is covered in the assembly of Section 6; it is not part of this theorem.

Proof. ABC has exactly two equal corner angles. Each corner angle is $m\alpha + k\frac{\pi}{3}$ (Section 2); since α/π is irrational, two corner angles are equal only if their pairs (m, k) coincide. So the corner-type triple of ABC has exactly one repeated type, and among the eleven triples of Section 2 exactly the two isosceles ones qualify. Hence the base angle is α or β. We use the invariant generated by the base angle, so that the equal sides land on $\frac{\pi}{3}$-coefficient 3 (odd) and pick up $f = -1$.

If the base angle is α, then $ABC = k(c, c, 2b + a)$ (equal sides kc, base $k(2b + a)$): the base is at direction 0 ($f_\alpha = +1$) and the equal sides at $\pi - \alpha = 3\frac{\pi}{3} - \alpha$ ($f_\alpha = -1$), so $\Phi_{f_\alpha}(\partial ABC) = k(2b + a) - 2kc = k(2b + a - 2c)$. The tiling count is $N = k^2(a + 2b)/b$; since N is prime and $\gcd(a + 2b, b) = 1$, we have $k^2 = b$, i.e. $k = \sqrt{b}$. By Corollary 36 and $c^2 = a^2 + ab + b^2$,

$$\frac{k(2b + a - 2c)}{c + a - b} = \frac{c - a - b}{k} \in \mathbb{Z}, \qquad k = \sqrt{b},$$

so $k \mid (a + b - c)$, contradicting Lemma 41. If the base angle is β, then $ABC = k(c, c, 2a + b)$, the equal sides are at $\pi - \beta$, and using f_β (for which the equal sides have $\frac{\pi}{3}$-coefficient 3) the same computation gives, with $k = \sqrt{a}$, $(c - a - b)/k \in \mathbb{Z}$, again contradicting Lemma 41 (the integer $a + b - c$ is symmetric in a, b). In both cases no tiling exists. ☐

Remark 44. The role of the two invariants is essential. For the base-angle-β shape, f_α places the equal sides at $\pi - \beta = \frac{2\pi}{3} + \alpha$, an *even* $\frac{\pi}{3}$-coefficient, so f_α gives an integer value and does not obstruct; f_β does. Each invariant is a necessary condition in its own right (Corollary 36), so using the one that fails is legitimate, not a choice of convenient frame: on any genuine tiling both invariants return an integer.

6. Resolution of the problem

Proof of Theorem 1. Let $p > 3$ be prime with $p \equiv 3 \pmod 4$ and $p \neq 3f^2 - e^2$ for every coprime $1 \le e < f$, and suppose some triangle is p-tiled. By the classification (Section 2) the tile is in one of the listed types. Consider the types other than a $2\pi/3$ tile on a non-equilateral triangle. For the $3\alpha + 2\beta$ branch, four of its five targets are excluded outright — the two scalene ones by the machine-checked cores of [3, Thms. 8, 12], the base-(α+β) target by Proposition 30 and the base-α target by Proposition 12 (the last two replacing [3, Thms. 18, 20], whose printed proofs are unsound). Its fifth target, base-β, is *not* excluded by any sound general argument (Remark 13); but its candidate counts are exactly the $3f^2 - e^2$ with $\gcd(e, f) = 1$, which the hypothesis excludes. The remaining branches exclude such a p by existing theorems [14, 2, 5, 4, 6]. For a $2\pi/3$ tile, Proposition 33 forces ABC isosceles and not equilateral, and Theorem 42 excludes that. Hence no such tiling exists. Finally $19 > 3$ is prime, $\equiv 3 \pmod 4$, and is not a base-β candidate: for $f \le 2$ one has $3f^2 - e^2 \le 11$; for $f = 3$, $3f^2 - e^2 \in \{26, 23, 18\}$; and for $f \ge 4$, $3f^2 - e^2 \ge 2f^2 + 2f - 1 \ge 39$. So no triangle is cut into 19 congruent triangles. □

7. Towards the full problem

Erdős's question asks for *all* N. Theorem 1 settles the negative side for primes; this section records what the present methods contribute to the full determination: a complete answer for primes, and, for each sporadic $2\pi/3$ branch, a determination of its *admissible spectrum* — the set of counts compatible with the area and invariant conditions.

Theorem 45 (Fibonacci families are the extremal ones)**.** *For a base-β family (e, f) with $\gcd(e, f) = 1$, $1 \le e < f$, write $M = f + e$ and $N_0 = 3f^2 - e^2$. Then*

$$M^2 - N_0 = -2\,(f^2 - ef - e^2),$$

so $|M^2 - N_0| \ge 2$, with equality if and only if $f^2 - ef - e^2 = \pm 1$, that is, if and only if (e, f) are consecutive Fibonacci numbers. In that case, writing $(e, f) = (F_n, F_{n+1})$,

$$M = F_{n+2}, \qquad N_0 = F_{n+2}^2 - 2(-1)^{n+1},$$

so the counts are exactly a Fibonacci square displaced by 2:

$$11 = 3^2{+}2, \quad 23 = 5^2{-}2, \quad 66 = 8^2{+}2, \quad 167 = 13^2{-}2, \quad 443 = 21^2{+}2, \quad 1154 = 34^2{-}2, \ \ldots$$

Proof. $(f + e)^2 - (3f^2 - e^2) = -2f^2 + 2ef + 2e^2 = -2(f^2 - ef - e^2)$. For coprime $e < f$ the integer $f^2 - ef - e^2$ is nonzero, whence $|M^2 - N_0| \ge 2$. Equality means $f^2 - ef - e^2 = \pm 1$, which is the Fibonacci characterisation: the solutions of $x^2 - xy - y^2 = \pm 1$ in positive integers are exactly the consecutive Fibonacci pairs. Then $M = F_n + F_{n+1} = F_{n+2}$ and $f^2 - ef - e^2 = (-1)^n$ give the displayed formula. □

Remark 46 (The field of computation)**.** For the tile $(ef, f^2 - e^2, f^2)$ one has $\sin\alpha = e\sqrt{D}/(2f^2)$ with $D = 4f^2 - e^2 = (2f - e)(2f + e)$, so a base-$\beta$ tiling's coordinates lie in $\mathbb{Q}(\sqrt{D})$. This is the field the exact search computes in; for $(e, f) = (1, 2)$ it is $\mathbb{Q}(\sqrt{15})$, the ring of all three tiling certificates.

Theorem 47 (Scaling ladder)**.** *Let a triangle T be cut into N copies of a tile t. Then for every integer $k \ge 1$ the scaled triangle kT is cut into k^2N copies of t.*

Proof. Subdivide kT by the triangular grid of side k: it decomposes into k^2 triangles, of which $\binom{k+1}{2}$ are translates of T and $\binom{k}{2}$ are its point-reflections, all congruent to T. Cut each into N copies of t. □

Corollary 48 (The realizable set of a base-β family is a union of multiples)**.** *Fix coprime $1 \le e < f$ and put $N_0 = 3f^2 - e^2$. Since the base-β target at scale m is $m \cdot (f^3, f^3, eN_0)$ — the same shape for every m — the set*

$$S(e,f) = \{\, m \ge 1 : \text{ the scale-}m\text{ target is cut into } m^2 N_0 \text{ copies of } (ef, f^2 - e^2, f^2) \,\}$$

satisfies $m \in S(e,f) \Rightarrow km \in S(e,f)$ for every $k \ge 1$. Hence $S(e,f)$ is the union of the multiples of its minimal elements, and $1 \in S(e,f)$ is not *implied by $S(e,f) \ne \emptyset$.*

Corollary 49 (An infinite family of tilings in an incommensurable branch)**.** *For $(e,f) = (1,2)$, tile $(2,3,4)$ and $N_0 = 11$: $1 \notin S$ (Table 1: the target $(8,8,11)$ is exhausted at 135 nodes), while $2 \in S$ and $3 \in S$ by the explicit 44- and 99-tilings. Therefore $S \supseteq 2\mathbb{Z}_{\ge 1} \cup 3\mathbb{Z}_{\ge 1}$, and*

$$N = 11m^2 \text{ is a number of congruent triangles for every } m \text{ divisible by } 2 \text{ or } 3: \quad 44,\ 99,\ 176,\ 396,\ 704,\ 891,\ 110$$

Moreover the 44- and 99-tilings are primitive*: neither arises from Theorem 47 applied to a smaller member of its own family, because that would require $1 \in S$. The values $N = 11m^2$ with $\gcd(m,6) = 1$ — the smallest being 275 — are undetermined.*

Remark 50. Two points. First, the exclusion of 11 together with the realizability of $11 \cdot 2^2$ and $11 \cdot 3^2$ shows that a base-β family can be empty at scale 1 and non-empty above it; the obstruction at $m = 1$ is genuinely a small-scale phenomenon and not a property of the family. Second, before this the smallest tiling known in any incommensurable branch had 1215 tiles; the corollary supplies infinitely many, of which 44 is the smallest. Whether $S(e,f) \ne \emptyset$ for every (e,f) — equivalently, whether every base-β family tiles at *some* scale — is open; the first untested instances are the scale-2 targets of the families $(2,3)$, $(3,4)$ and $(1,4)$, with $N = 92$, 156 and 188.

Theorem 51 (The prime case)**.** *Let p be a prime with $p \not\equiv 11 \pmod{12}$. Then p is a number of congruent triangles into which some triangle can be cut if and only if*

$$p = 2, \qquad p = 3, \qquad \text{or} \qquad p \equiv 1 \pmod 4.$$

The same equivalence holds for those $p \equiv 11 \pmod{12}$ that have been settled by exhaustive search, currently $11, 23, 47, 59, 71, 107$ (Table 1); for the remaining primes $\equiv 11 \pmod{12}$ — the smallest being 83 — the "only if" direction is open.

Remark 52 (The prime case of Erdős's problem, in one line)**.** Combining Theorem 4 with the branch analysis, the prime part of the problem has been reduced to a single congruence class. A prime p lies in the spectrum if $p = 2$, $p = 3$ or $p \equiv 1 \pmod 4$ (classical constructions), and does not if $p \equiv 7 \pmod{12}$ (Corollary 7); the class $3 \pmod{12}$ contains only $p = 3$. What remains is exactly

$$p \equiv 11 \pmod{12},$$

which is precisely the set of base-β candidates, conjecturally empty of tile counts, and verified so for $11, 23, 47, 59, 71, 107$. By Dirichlet the resolved and unresolved halves of the primes $\equiv 3 \pmod 4$ have equal density, so the prime problem is settled for a set of primes of density $\frac{3}{4}$ among all primes, and open on a set of density $\frac{1}{4}$.

Proof. If. For $p = 2$, the altitude from the right angle splits an isosceles right triangle into two congruent triangles. For $p = 3$, join the centre of an equilateral triangle to its vertices. For $p \equiv 1 \pmod 4$, Fermat's two-squares theorem gives $p = e^2 + f^2$ with $e, f \ge 1$; splitting a right triangle with legs e, f by the altitude from the right angle and tiling the two similar pieces quadratically with e^2 and f^2 congruent copies realizes $N = e^2 + f^2$ (the biquadratic tiling [15, 4]). *Only if.* Theorem 1, whose dependencies this direction inherits. ☐

Theorem 53 (Isosceles admissible spectrum)**.** *Let (a, b, c) be a primitive $120°$-triple and write $b = de^2$ with d squarefree. Every N-tiling of the base-α isosceles target has scale $k = dew$ for a positive integer w, and*

$$N = d\,w^2\,(a + 2b), \qquad e \mid w\,(c - a - b), \qquad \frac{w(c - a - b)}{e} \equiv N \pmod 2.$$

Symmetrically, with $a = d'e'^2$, the base-β target forces $N = d'w^2(2a + b)$, $e' \mid w(c - a - b)$ and the same parity for $w(c - a - b)/e'$.

Proof. By the area identity $Nb = k^2(a+2b)$ and $\gcd(a+2b, b) = 1$ we get $b \mid k^2$. For a prime $p \mid d$ the valuation $v_p(b) = 2v_p(e) + 1 \le 2v_p(k)$ forces $v_p(k) \ge v_p(e) + 1$, and for $p \nmid d$, $v_p(k) \ge v_p(e)$; hence $de \mid k$, say $k = dew$, and substituting back, $N = dw^2(a + 2b)$. By Corollary 36 and the identity $(c - a - b)(c + a - b) = b(2b + a - 2c)$, the invariant value is $M = k(c - a - b)/b$, so $b \mid k(c - a - b)$, i.e. $de^2 \mid dew(c - a - b)$, i.e. $e \mid w(c - a - b)$; and $M = w(c - a - b)/e \equiv N \pmod 2$ by Corollary 36. ☐

Theorem 42 is the degenerate case: N prime forces $d = w = 1$, so $b = e^2$ and $e \mid (c - a - b)$, which Lemma 41 refutes. The residual divisibility is governed by the following classification.

Proposition 54 (Solvability of $e \mid (a + b - c)$)**.** *Let $b = de^2$, d squarefree, $\gcd(a, b) = 1$. Then $e \mid (a+b-c)$ holds for a primitive $120°$-triple if and only if there is an integer j with $d < j < 2d$ such that ej is even,*

$$a = \frac{e^2(2d - j)(2d + j)}{4(j - d)}$$

is a positive integer coprime to de^2; in that case $c = a + \frac{e^2 j}{2}$. In particular, for $d = 1$ the range $(1, 2)$ contains no integer, which re-proves Lemma 41; for $d = 2$, $j = 3$, $e = 2$ one obtains $(a, b, c) = (7, 8, 13)$.

Proof. Since $e \mid b$, the hypothesis gives $c \equiv a \pmod e$, and $0 < c - a < b$ writes $c = a + et$ with $1 \le t \le de-1$. Substituting into $c^2 = a^2+ab+b^2$ and dividing by e, $a(2t-de) = e(de-t)(de+t)$. As $e \mid a(2t - de)$ and $\gcd(a, e) = 1$, we get $e \mid 2t$, say $2t = ej$ with $1 \le j \le 2d - 1$; substituting $t = ej/2$ gives $4a(j - d) = e^2(2d - j)(2d + j)$, whose right side is positive, forcing $j > d$ and the displayed formula. Conversely, given such j, setting $t = ej/2$, a as displayed and $c = a + et$ reverses the computation: $c^2 = a^2 + ab + b^2$ and $a + b - c = e(de - t)$ is divisible by e. ☐

Theorem 55 (Spectrum lattice)**.** *Let (a, b, c) be a primitive $120°$-triple, $b = de^2$ with d squarefree, $r = c - a - b$, $g = \gcd(e, r)$, $e_1 = e/g$, and*

$$T \;=\; \frac{r}{g} + d\,e_1^2\,(a + 2b) \pmod 2.$$

Set $E = e_1$ if T is even and $E = 2e_1$ if T is odd. Then the counts admissible for the base-α isosceles target (Theorem 53 with the parity clause) are exactly

$$N = d\,(Eu)^2\,(a + 2b), \qquad u = 1, 2, 3, \ldots$$

Proof. By Theorem 53 the conditions on w are $e \mid wr$ and $wr/e \equiv N \pmod 2$. As $\gcd(e_1, r/g) = 1$, the divisibility is equivalent to $e_1 \mid w$; writing $w = e_1u$, the count is $M = ur/g$, and using $u^2 \equiv u \pmod 2$,

$$M - N = \frac{ur}{g} - d\,e_1^2u^2(a + 2b) \equiv u\Big(\frac{r}{g} + d\,e_1^2(a + 2b)\Big) = uT \pmod 2,$$

so the parity clause holds for all u when T is even, and exactly for even u when T is odd. $\square$

Theorem 55 uses only the base-aligned invariant. The mirror invariant is also a necessary condition on the same target, and it closes the spectrum completely.

Theorem 56 (Spectrum theorem)**.** *Let (a, b, c) be a primitive $120°$-triple. The tile counts satisfying* all *the invariant conditions on the base-α isosceles target — integrality and parity of both M_α and M_β, together with the area equation — are exactly Zhang's family*

$$N \;=\; m^2\, b\,(a + 2b), \qquad m = 1, 2, 3, \dots$$

Symmetrically, the base-β target admits exactly $N = m^2 a(2a + b)$, and the F_1 target exactly $N = m^2 b(a + b)$.

Proof. Write $X = c + a - b$, $Y = c + b - a$, so $XY = 3ab$ by (1). On the base-α target with scale k, $M_\alpha = k(2b + a - 2c)/X$ and $M_\beta = k(2b + a + 2c)/Y$ (the equal sides carry $\frac{\pi}{3}$-coefficient 3 in the α-frame and 2 in the β-frame). Since $(2b + a - 2c) + 2X = 3a$ and $(2b + a + 2c) - 2Y = 3a$, the two integralities are equivalent to

$$X \mid 3ak \qquad \text{and} \qquad Y \mid 3ak,$$

and, setting $A = 3ak/X$, $B = 3ak/Y$, the parity clauses of Corollary 36 read $A \equiv B \equiv N \pmod 2$. Recall $k = dew$ and $N = dw^2(a + 2b)$, with $b = de^2$.

If $e \mid w$, say $w = em$, then $k = de^2 m = bm$, so $A = Ym$ and $B = Xm$ are integers, and $A - N \equiv m\big(Y + b(a + 2b)\big) \pmod 2$ using $m^2 \equiv m$; now $Y + b(a + 2b) \equiv c + a + b + ab \equiv 0 \pmod 2$, because (1) gives $c \equiv a + b + ab$; likewise for B. So every member of Zhang's family is admissible.

Conversely, fix a prime p and let $x, y, \sigma, \varepsilon, \delta, \omega$ be the p-adic valuations of $X, Y, 3a, e, d, w$; then $x + y = \sigma + \delta + 2\varepsilon$ and the divisibilities give $\max(x, y) \le \sigma + \delta + \varepsilon + \omega$, so $\omega \ge \varepsilon - \min(x, y)$. A common prime of X and Y divides $X + Y = 2c$ and $Y - X = 2(b - a)$; for odd p this forces $p \mid c$ and $p \mid b - a$, whence $3 \mid c$ or $p \mid \gcd(a, b)$ — both impossible ($3 \nmid c$ as in Proposition 33). So $\min(x, y) = 0$ for every odd p, and $\omega \ge \varepsilon$ there. At $p = 2$: if ab is odd then X, Y are odd and the same argument applies. If b is even (so a is odd), a two-line computation modulo 8 shows $v_2(b) \notin \{1, 2\}$: $b \equiv 2 \pmod 4$ gives $c^2 \equiv 1 + 2 \equiv 3 \pmod 4$, and $v_2(b) = 2$ gives $c^2 \equiv 1 + 4 \equiv 5 \pmod 8$, both impossible. So $v_2(b) \ge 3$. Here $v_2(X + Y) = v_2(2c) = 1$ forces $\min(x, y) = 1$, and the divisibilities alone allow $\omega = \varepsilon - 1$; but then exactly one of A, B is odd (their 2-valuations are 0 and $v_2(b) - 2$), while $v_2(N) = \delta_2 + 2\omega = v_2(b) - 2 \ge 1$ makes N even — so the parity clause fails, and $\omega \ge \varepsilon$ at $p = 2$ as well. Hence $e \mid w$, i.e. $N = m^2 b(a + 2b)$. The base-β case is the mirror $a \leftrightarrow b$. For F_1 the counts are $M_\alpha = 3ak/X - k$ and $M_\beta = 3ak/Y + k$ (Section 7), so the same divisibilities appear, the parity clauses read $A \equiv B \equiv N + k \pmod 2$, and in the boundary case $v_2(k) = \delta_2 + 2\varepsilon_2 - 1 \ge 2$ makes $N + k$ even against an odd count; the argument is otherwise identical, giving $N = m^2 b(a + b)$. $\square$

Remark 57. The single-invariant lattice of Theorem 55 can be strictly coarser than $e\mathbb{Z}$ (the smallest example is $(11, 24, 31)$, whose lattice admits $N = 354$), so the single invariant alone does not confine the admissible counts to Zhang's constructions. The mirror invariant of Theorem 56 eliminates every such value (354 by parity, 15689 on $(19, 261, 271)$ already by integrality), and the fully refined spectrum *coincides with Zhang's family on every tile* — verified exhaustively over all 138 primitive triples with $a, b < 400$. The necessary side of the isosceles and F_1 branches is therefore closed. What remains open in these branches is exactly the realizability of the *small* family members. Not all of them are open: a family member that is also a sum of two squares (or 2, 3, 6 times a square) is realized by the commensurable construction [5], so the genuinely undecided members are precisely those with a prime $\equiv 3 \pmod 4$ to an odd power. (Thus

$N = 40 = b(a+b)$ on $(3,5,7)$ is realized as 2^2+6^2, and $N = 65 = b(a+2b)$ on $(3,5,7)$ as 1^2+8^2, while $105 = 3\cdot 5\cdot 7$ is not a sum of two squares.) Zhang's constructions cover m in certain residue classes from some bound onward, so the sharpest open instances are the smallest such members: $N = 105 = b(a+b) = 7\cdot 15$ on the F_1 target $(105, 56, 91)$ of the tile $(8, 7, 13)$, then 120, 132, 154, 184. Each has membership undecided by any known necessary condition, and is exactly the kind of bounded existence question the engine of Section 7 addresses — though for N of this size a complete exhaustive search is not yet computationally in reach. We note the scope boundary explicitly: the construction families of [9] concern the six sporadic targets whose tile carries a $2\pi/3$ angle (plus a $\pi/3$ variant); the $3\alpha + 2\beta = \pi$ branches carry no constructed family in the literature known to us. There the situation is now reversed: the base-β member $(1,2)$ is settled with *no* residue class and *no* lower bound (Theorem 5), and the collar induction behind it reduces every base-β member to three finite seed searches (companion note, realizability section).

Remark 58. Admissibility is necessary, not sufficient. The smallest admissible value is $N = 33 = 3\cdot 11$ on the tile $(5,3,7)$ (where $d = 3$, $e = 1$, so the divisibility is vacuous); it is nevertheless excluded by Beeson's boundary analysis, which rules out $N < 36$ [4], while Herdt's 2673-tile tiling is the member $w = 9$ of the same family $33w^2$. The parity clause is decisive: on the tile $(7, 8, 13)$ (where $d = 2$, $e = 2$, $c - a - b = -2$) the divisibility admits all $N = 2w^2 \cdot 23$, but the count $M = -w$ must have the parity of $N = 2w^2\cdot 23$, which is even — forcing w even, say $w = 2m$; the refined spectrum is exactly $N = 184m^2 = m^2\, b(a + 2b)$ — *precisely* Zhang's constructed family [9] for this tile. In particular $N = 46$, previously the smallest value passing every known necessary condition on this branch, is *excluded* (Section 7 checks the remaining branches), in exact agreement with the prediction of Zhang's conjecture. Zhang's constructions realize $N = m^2 b(a+2b)$ for m in certain residue classes; the refined spectrum bounds the realized set from outside, and Theorem 55 determines it exactly.

Proposition 59 (The rationality input, partly internalised)**.** *On the six shapes whose invariant-product constant is $\kappa = -a/(a + 2b)$ or $-b/(2a + b)$* (iso base-α, iso base-β, F_3, F_4, F_3', F_4')*, the tile's side ratios are rational, provided the configuration* $(\star)$ *below does not occur. No appeal to* [8] *is made.*

Proof. The counts M_α, M_β and the tile count N are integers, so $\kappa = M_\alpha M_\beta/N$ is rational. With $P = M_\alpha M_\beta$, the identity $P(a + 2b) = -aN$ clears to $a(N + P) = -2bP$, an integer-coefficient linear relation; the coefficient is nonzero because $P(a + 2b) = -aN < 0$ forces $P < 0$. Hence $a/b = -2P/(N + P) \in \mathbb{Q}$, and likewise $b/a \in \mathbb{Q}$ in the mirrored family.

For the third ratio, use the tiling. Each of the six targets has a side that is a rational multiple λc of c and a side that is rational in a, b; up to a common scale the side data are $(c, c, a + 2b)$, $(c, c, 2a + b)$, $(c^2, c(a + 2b), 3b(a + b))$, $(ac, b(2a + b), c(a + b))$, $(c^2, c(2a + b), 3a(a + b))$ and $(bc, a(a + 2b), c(a + b))$. A side is covered by whole tile edges, so it carries a walk $Pa + Qb + Rc$ with $P, Q, R \geq 0$. On a c-carrying side this gives $(\lambda - R)c = Pa + Qb$, exhibiting c as a rational combination of a and b unless $\lambda = R$, in which case $Pa + Qb = 0$ and hence $P = Q = 0$: the side consists of c-edges alone. On a side rational in a, b it gives $Rc \in \mathbb{Q}(a, b)$, so a single c-edge suffices. The only surviving configuration is

$(\star)$ every c-carrying side is covered by c-edges alone, and every other side by a- and b-edges alone,

and outside it $c/b \in \mathbb{Q}$, so all three ratios are rational. □

Remark 60. The hypothesis is explicit and strictly weaker than the imported theorem: excluding $(\star)$ is a corner-and-walk exclusion of the kind carried out for the base-β target, where the γ-trap shows every side carries a c-edge. It is not established here for these six shapes. The remaining two shapes are unaffected: on F_2 the same elimination gives a quadratic in a/b with discriminant $9\kappa^2 - 30\kappa + 9$, and on the equilateral target $\kappa = 3$ is constant, so both continue to use [8]. The arithmetic is machine-checked in `lean/RationalityFree.lean`. That the condition has content is visible from the fact that $a/b \in \mathbb{Q}$ alone does not give $c/b \in \mathbb{Q}$: the latter asks that $(a/b)^2 + (a/b) + 1$ be a rational square, which holds for $a/b = 3/5$ and $7/8$ but not for $a/b = 1$ or 2; the tiles for which it holds are the integer triangles with a $2\pi/3$ angle.

Remark 61 (The parallelogram of [9] is sharp)**.** The constructions of [9] pass through the parallelogram $Q(y, x)$ with angles $2\pi/3, \pi/3$ and sides $y = ab$, x, which is assembled there from sub-parallelograms of sides (ab, a) and (ab, b); the assembly therefore needs $x \in \langle a, b\rangle$, and the hypothesis $x > ab-a-b$ of that lemma is exactly the Frobenius number of $\langle a, b\rangle$. The undecided small members of the constructed families are the ones whose parameter falls in the Frobenius gap, so it is natural to ask whether $Q(ab, x)$ might still be tileable there by some assembly that is not a lattice of those sub-parallelograms.

It is not. In this branch the only two-tile parallelogram carrying the angles $2\pi/3, \pi/3$ is the one glued along c (sides a, b); gluing along a or b gives angles $(\beta + \gamma, \alpha)$ resp. $(\alpha + \gamma, \beta)$, which do not match. A lattice of the admissible brick reproduces exactly $x = ka + lb$. Since $Q(ab, x)$ contains exactly $2x$ tiles, the gap cases are small enough to decide outright, and exhaustive search gives no tiling: on the tile $(3, 5, 7)$ the values $x = 4$ and $x = 7$ return NO TILING (1 and 140 nodes) while the control $x = 8 = 3 + 5$ is found in 62 nodes, and on $(7, 8, 13)$ the value $x = 13$ returns NO TILING (3967 nodes) while the control $x = 15 = 7+8$ is found in 667. The two failing values $x = c$ are instructive: there the boundary walk exists — a single c-edge spans the side — so the obstruction is not the walk arithmetic but the interior. We record the conclusion at the strength the evidence supports: for the instances tested, $Q(ab, x)$ is tileable if and only if $x \in \langle a, b\rangle$, so the hypothesis of that lemma is necessary and not merely convenient; in general this is a conjecture.

This closes one route to the small members but not the question itself: a target triangle with N below the constructed range could still be tiled by a dissection that never uses $Q(ab, x)$. That question can be attacked directly, one instance at a time.

Remark 62 (Instances below the constructed range)**.** The families of [9] realize $N = m^2\kappa$ for m from an explicit bound onward, typically $m \geq M = 3\lceil (c^2-a-b)/(ab)\rceil$, and the conjecture there is that no smaller m occurs. The instances with $m < M$ are small enough to decide outright. Exhaustive search settles the following, each returning NO TILING; the parameters are those of [9, Table 1], and $M = 9$ in every row listed.

tile	target	m	N	nodes
$(3, 5, 7)$	equilateral m^2ab	1	15	156
$(3, 5, 7)$	equilateral	2	60	853 731
$(7, 8, 13)$	equilateral	1	56	156 646
$(5, 16, 19)$	equilateral	1	80	1 222 714
$(3, 5, 7)$	$(\beta, \alpha+\beta, 2\alpha+\beta)$ $m^2a(a + b)$	1	24	8 658
$(3, 5, 7)$	$(\beta, \beta, \pi - 2\beta)$ $m^2a(b + 2a)$	1	33	171 950

The targets are determined by the tile: for $(3, 5, 7)$ one has $\cos\alpha = 13/14$ and $\alpha + \beta = \pi/3$, whence $\sin(2\alpha + \beta) = \sin(\alpha + 2\beta) = 4\sqrt{3}/7$, $\sin 2\beta = 55\sqrt{3}/98$ and $\sin 2\alpha = 39\sqrt{3}/98$; the rows above are then the triangles $(15, 21, 24)$ and $(21, 21, 33)$, whose areas are exactly 24 and 33 tile areas. This is evidence for the conjecture of [9] on six instances spanning three tiles and two target shapes, not a proof of it: the conjecture asserts the exclusion for all $m < M$ on all six sporadic targets. We note also that $N = 24$ and $N = 33$ do occur elsewhere in the spectrum, by commensurable constructions; the statement decided here is per (target, tile) pair.

Proposition 63 (The other branches)**.** *With the same normalizations: F_1 forces $N = dw^2(a+b)$ where $b = de^2$, d squarefree (from $b \mid k^2$ as above); F_2, F_3, F_4 force $N = N_0 k^2$ exactly, with $N_0 = (a+2b)(2a+b)$, $3(a+b)(a+2b)$, $(a+b)(2a+b)$; the tile-similar shape forces $N = n^2$; and an equilateral target of side S forces $ab \mid S^2$, $N = S^2/(ab)$ (so N and ab have the same squarefree kernel) together with $(c+a-b) \mid 3S$ and $(c+b-a) \mid 3S$.*

Proof. The scalene statements are the area identity (5) with the side-vectors of Section 3 (F_1 repeats the $b \mid k^2$ analysis). For the equilateral target the three sides, traversed counterclockwise, differ in direction by $\frac{2\pi}{3} = 2 \cdot \frac{\pi}{3}$, so f takes one sign on all three and $\Phi_f(\partial ABC) = \pm 3S$ for both invariants; Corollary 36 gives the two divisibilities, and the area identity gives $N\,ab = S^2$. □

The equilateral $2\pi/3$ spectrum. The equilateral divisibilities of Proposition 63 reduce to a clean finite criterion.

Proposition 64 (Equilateral admissibility)**.** *Let an equilateral triangle of side S be N-tiled by a $2\pi/3$ tile (a, b, c), and put $X = c + a - b$, $Y = c + b - a$. Then $XY = 3ab$, and $s := 3S/X$, $t := 3S/Y$ are positive integers with*

$$st = 3N, \qquad s \equiv t \equiv N \pmod 2, \qquad (t-s)^2 + 16N = q^2 \ \textit{ for an integer } q;$$

conversely (s, t, q) determine the instance: (a, b, c) is proportional to $\big(q+(t-s),\ q-(t-s),\ 2(s+t)\big)$, and primitivity pins the scale, hence S.

Proof. $XY = c^2 - (a-b)^2 = 3ab$ by (1). By Proposition 63, $X \mid 3S$ and $Y \mid 3S$, so $s, t \in \mathbb{Z}_{\geq 1}$; indeed $s = M_\alpha$ and $t = M_\beta$ are the two invariant counts, so $s \equiv t \equiv N \pmod 2$ by Corollary 36, and $st = 9S^2/(XY) = 9\,Nab/(3ab) = 3N$. From $X - Y = 2(a-b)$ we get $a - b = \frac{3S}{2}\big(\frac{1}{s} - \frac{1}{t}\big) = \frac{S(t-s)}{2N}$, and $4ab = 4S^2/N$ gives $(a+b)^2 = (a-b)^2 + 4ab = \frac{S^2}{4N^2}\big[(t-s)^2 + 16N\big]$; as $a+b$ is an integer, $q := 2N(a+b)/S$ is a rational with integer square, hence an integer, and $(t-s)^2 + 16N = q^2$. Then $a - b$, $a + b$ and $c = (X+Y)/2 = \frac{S(s+t)}{2N}$ give the displayed proportions. □

The equilateral criteria as divisor conditions. Both equilateral square criteria — ours and Beeson's — complete the square.

Proposition 65 (Conic form)**.** *For an equilateral target: a $2\pi/3$ tile requires a factorization $uv = 16N^2$ with $u+v \equiv 0 \pmod 2$ and $\frac{u+v}{2} - 5N = s^2$ for a divisor s of $3N$ with $s \equiv 3N/s \equiv N \pmod 2$; a $\pi/3$ tile requires a factorization $uv = 16N^2$ with $\frac{u+v}{2} = 5N - M^2$ for some $0 < M < \sqrt{N}$, $M \equiv N \pmod 2$. These are equivalent to the criteria of Proposition 64 and of* [6, Thm. 3] *respectively.*

Proof. For the $2\pi/3$ case, with $t = 3N/s$ the condition $(t-s)^2 + 16N = q^2$ becomes, after multiplying by s^2, $(qs)^2 = (3N - s^2)^2 + 16Ns^2 = (s^2+N)(s^2+9N)$, i.e. $(s^2+5N)^2 - (qs)^2 = 16N^2$; factor the difference of squares. For the $\pi/3$ case, $(9N - M^2)(N - M^2) = y^2$ is $(5N - M^2)^2 - 16N^2 = y^2$, i.e. $(5N - M^2 - y)(5N - M^2 + y) = 16N^2$. □

The necessary side of the equilateral branch is thus elementary — a divisor condition on $16N^2$ — and, per N, pins finitely many fully determined instances. Both square-completions of Proposition 65 are formalized and machine-checked in Lean (`lean/EquilateralConic.lean`): the elimination of t giving $(qs)^2 = (s^2+N)(s^2+9N)$, the resulting factorization $(s^2 + 5N - qs)(s^2 + 5N + qs) = 16N^2$, and the $\pi/3$ identity $(5N - M^2)^2 - 16N^2 = (9N - M^2)(N - M^2)$, all over $\mathbb{Z}$. The elliptic curves of Laczkovich [13] concern the *realizability* of such instances, not their enumeration.

The values 14 and 15. The machinery above, combined with the published branch theorems, decides every branch of the classification for $N = 14$ and $N = 15$ by finite computations (`verify_frontier.py`): the commensurable, tile-similar, right-angle and $\gamma = 2\alpha$ branches exclude both values by their known N-forms; the sporadic $2\pi/3$ shapes exclude both by the admissible spectra of Theorems 53 and Proposition 63; the $\pi/3$-tile equilateral case excludes both by Beeson's square criterion ($(9N - M^2)(N - M^2)$ is a perfect square for some $M < \sqrt{N}$ [6, Thm. 3], which fails for every admissible M); and of the five target shapes of the $3\alpha + 2\beta = \pi$ branch [3], four exclude both values through their published tiling equations (checked by exact rational-root enumeration), while the isosceles target with base angles $\alpha+\beta$ admits, for $N = 14$, the single candidate tile $(45, 56, 81)$ with $M = 2$ — which is destroyed by Beeson's boundary congruence $M \equiv -m \pmod{\gcd(a, c)}$ (it forces at least seven b-edges of length 56 on a base of length 140).

What survives everything is four fully explicit finite instances:

N	branch	tile	target
14	equilateral, $2\pi/3$	$(7, 8, 13)$	equilateral, $S = 28$
15	equilateral, $2\pi/3$	$(3, 5, 7)$	equilateral, $S = 15$
15	$3\alpha + 2\beta = \pi$, iso-$(\alpha+\beta)$, $M = 1$	$(56, 15, 64)$	$(120, 120, 105)$
15	$3\alpha + 2\beta = \pi$, iso-$(\alpha+\beta)$, $M = 3$	$(4, 15, 16)$	$(60, 60, 15)$

In each, the tile is *uniquely determined* (Proposition 64 for the equilateral rows; the tiling equation $N/M^2 = (1 + s)/(1 - s)$ of [3, Thm. 17] for the isosceles rows), so each is a single bounded existence question: can 14 (resp. 15) congruent copies of the stated tile fill the stated triangle?

Each instance was settled by an exhaustive, exact search (`code/engine/`). The engine is an advancing-front algorithm over $\mathbb{Q}(\sqrt{D})$ ($D = 3, 23, 7$; every geometric decision exact, no floating point) whose branching rule is provably complete: at the lexicographically least vertex v of the unfilled region — necessarily convex — every tile of any tiling whose closure contains v has a corner there (an edge through v would subtend angle π), so the clockwise-most of them has a corner at v with one side along the clockwise boundary ray; this leaves at most six placements (three corner types, two adjacent sides each, the two choices being mirror images). An exhausted search is therefore a proof of non-existence. Two prunes are used, each a theorem about completable states: the area of every region component is a positive integer multiple of the tile area; and a maximal straight boundary run with *convex* endpoints has length in $\mathbb{N}a+\mathbb{N}b+\mathbb{N}c$ (whole tile edges cannot pass a convex corner — runs with a reflex endpoint are not pruned, as an edge may continue past a reflex corner into the region). The engine passes positive controls — it finds the $N = 4$ reptile subdivisions in both tile families and a hand-built non-edge-to-edge two-tile configuration with T-junctions, each found tiling re-verified by an independent checker — and, as a robustness check, all four verdicts below are unchanged when the run prune is disabled entirely (node counts without it in parentheses).

instance	tile	target	verdict	nodes
$N = 14$, equilateral	$(7, 8, 13)$	$S = 28$	exhausted, no tiling	133 (482)
$N = 15$, equilateral	$(3, 5, 7)$	$S = 15$	exhausted, no tiling	156 (604)
$N = 15$, iso-$(\alpha+\beta)$	$(56, 15, 64)$	$(120, 120, 105)$	exhausted, no tiling	37 (526)
$N = 15$, iso-$(\alpha+\beta)$	$(4, 15, 16)$	$(60, 60, 15)$	exhausted, no tiling	2 (10)

Theorem 66. *No triangle can be cut into* 14 *congruent triangles, and none into* 15*.*

Proof. By the branch sweep above, any 14- or 15-tiling would live in one of the four listed instances, each with its tile and target uniquely determined; the exhaustive searches refute all four. The dependencies are those of Theorem 1 together with [6, Thm. 3] and the declared computer assistance (exact arithmetic, complete branching, independently validated). From [3] we use only the *tiling equations* of Theorems 3, 7, 11, 14, 17 and 19 — that is, the Diophantine relations each derives between the tile and its target — and *not* their $g \mid M$ conclusions. This distinction is not cosmetic: Remark 21 shows that Theorem 14's conclusion is false, and Remark 11 that Theorem 19's proof is unsound. The equations we use are re-derived citation-free in Remark 29, so the present theorem does not rest on the defective steps. □

The values 21, 22, 30, and the instance 46. The same sweep, sharpened by the parity clause of Corollary 36, extends further (`verify_frontier.py`). Three remarks carry the new weight. First, the invariant applies to the *scalene* F_1 target: its sides $k(a+b), ka, kc$ have $\frac{\pi}{3}$-coefficients $0, 2, 3$ (the same turning computation as Lemma 35), so

$$M_\alpha = \frac{k(2a+b-c)}{c+a-b}, \qquad M_\beta = \frac{k(2a+b+c)}{c+b-a}$$

must be integers of the parity of N. For $N = 21$ this destroys the unique structural candidate, the genuine tile $(5, 16, 19)$ with $k = 4$: $M_\alpha = 28/8 \notin \mathbb{Z}$ — while Zhang's constructed F_1 family $N = m^2 b(a + b)$ passes, so the invariant separates the candidate from the constructions. (For $N = 30$ the candidate $(7, 8, 13)$, $k = 4$ has $M_\alpha = 3$, killed by parity.) Second, for $N = 22$ and $N = 30$ *every* branch dies arithmetically — the two isosceles-$(\alpha+\beta)$ candidates of 22 fall to the boundary congruence, those of 30 likewise — so no search is needed at all. Third, for $N = 21$ exactly two instances survive, both isosceles-$(\alpha+\beta)$: $M = 1$, tile $(110, 21, 121)$ on $(231, 231, 210)$, and $M = 3$, tile $(10, 21, 25)$ on $(105, 105, 42)$; both live in $\mathbb{Q}(\sqrt{6})$, and the engine exhausts both:

instance	tile	target	verdict	nodes
N=21, iso-$(\alpha+\beta)$, M=1	$(110, 21, 121)$	$(231, 231, 210)$	exhausted	107 (4266)
N=21, iso-$(\alpha+\beta)$, M=3	$(10, 21, 25)$	$(105, 105, 42)$	exhausted	15 (102)

Finally, the parity clause settles the value that Remark 58 identified as the sharpest test of Zhang's conjecture: $N = 46$ passes the divisibility spectrum on $(7, 8, 13)$ but not the parity, and the sweep excludes every other branch, so 46 is not realizable — exactly as Zhang's conjecture predicts.

The sweep continues: $N = 38$ and $N = 42$ die in every branch by arithmetic alone (both squarefree, so the $\gamma = 2\alpha$ branch is closed by [4, Thm. 11.7], and no sporadic, equilateral or $3\alpha + 2\beta$ candidate survives), while 33, 35 and 39 each reduce to two isosceles-$(\alpha+\beta)$ instances (the sporadic-$2\pi/3$ candidate $(5, 3, 7)$ at $N = 33$ is excluded by Beeson's $N < 36$ boundary analysis [4]); all six searches exhaust:

instance	tile	target	nodes
N=33, M=1	$(272, 33, 289)$	$(561, 561, 528)$	489
N=33, M=3	$(28, 33, 49)$	$(231, 231, 132)$	49
N=35, M=1	$(306, 35, 324)$	$(630, 630, 595)$	523
N=35, M=5	$(6, 35, 36)$	$(210, 210, 35)$	2
N=39, M=1	$(380, 39, 400)$	$(780, 780, 741)$	853
N=39, M=3	$(40, 39, 64)$	$(312, 312, 195)$	67

Theorem 67. *No triangle can be cut into 21, 22, 30, 33, 35, 38, 39, 42, or 46 congruent triangles.*

Proof. The sweeps above; for 21, 33, 35 and 39, additionally the exhaustive searches listed. Dependencies as in Theorem 66. □

$N = 44$ **is realizable.** The value 44, the smallest undetermined after Theorem 67, has three live branches. Two die: the isosceles-$(\alpha+\beta)$ instance (tile $(30, 11, 36)$, target $(132, 132, 110)$, from Theorem 17 of [3] at $M = 2$) is exhausted by the engine (15654 nodes, independently in both engine implementations), and the $\gamma = 2\alpha$ branch is excluded by Beeson's Theorem 11.18 [4] ($N \geq 45$ for that branch; his Table 3 lists and eliminates the unique $N = 44$ boundary candidate $(25, 11, 30)$). The third branch is the isosceles-base-β case of [3, Thm. 14]: at $(N, M) = (44, 6)$ the tiling equation gives $s = a/c = \frac{1}{2}$, hence the tile $(2, 3, 4)$ — angles $(\alpha, \beta, 2\alpha+\beta)$ with $3\alpha + 2\beta = \pi$, $\cos\alpha = \frac{7}{8}$, $\cos\beta = \frac{11}{16}$, $\cos\gamma = -\frac{1}{4}$ — and the target $(16, 16, 22)$ with base angles β. Here the exhaustive search does not exhaust: *it finds a tiling*, at node 858163.

Theorem 68. *A triangle can be cut into* 44 *congruent triangles: the isosceles triangle with sides* $(16, 16, 22)$ *is tiled by* 44 *congruent copies of the* $(2, 3, 4)$ *triangle (Fig. 1). Consequently* $44m^2$ *is a tile count for every* $m \geq 1$.

Proof. The tiling is exhibited explicitly (`code/engine/tiling_N44B.txt`: exact coordinates in $\mathbb{Q}(\sqrt{15})$), and its certificate is *machine-verified in Lean 4 with no axioms whatsoever* (`lean/Tiling44.lean`: the kernel checks, by pure computation over $\mathbb{Z}[\sqrt{15}]$ after scaling by 8, that all 44 triangles have squared side multiset exactly $\{256, 576, 1024\} = (8 \cdot (2, 3, 4))^2$, are positively oriented, lie in the closed target via half-plane checks, admit an explicit separating edge-line for each of the 946 pairs, and have signed areas summing exactly to the target's; `#print axioms` reports the theorem depends on no axioms). The bridge from the certificate to the tiling statement is classical: closed half-plane separation gives pairwise disjoint interiors (convexity), the vertices-in-target checks give containment (convexity of the target), and then exact equality of total area forces the union of the closed tiles to be the whole target, since the complement is open with measure zero. So this is an N-tiling in the strict sense — *unconditionally*: this theorem depends on none of the cited classification inputs. The consequence follows by subdividing each tile into m^2 similar copies. $\square$

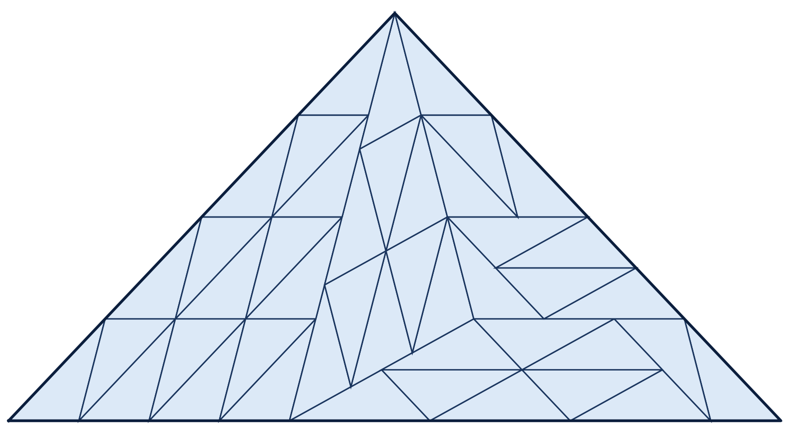

FIGURE 1. The 44-tiling of $(16, 16, 22)$ by the tile $(2, 3, 4)$, found by exhaustive search and verified exactly. To our knowledge the first tiling in any of the three isosceles $3\alpha + 2\beta = \pi$ cases, and the smallest known triangle tiling in any incommensurable branch (the previously smallest was Herdt's 1215).

Three remarks. First, 44 is not a square, not a sum of two squares ($11 \equiv 3 \bmod 4$), and not $2, 3, 6$ times a square: it is the first tile count *outside the commensurable families* to be certified realizable at small size, and the first new member of the tile-count set discovered by these methods — every other frontier value was excluded. Second, Beeson's constructions in [3] cover the two scalene $3\alpha + 2\beta = \pi$ shapes (his triquadratic tilings, smallest $N = 28$ — which, we note, settles 28's membership and is used in our accounting below); the isosceles cases had, to our knowledge, no known tiling, and the smallest known tiling in any incommensurable branch was Herdt's 1215. Third, the value demonstrates that the small members of the $3\alpha + 2\beta = \pi$ tables are not uniformly impossible: each must be searched, and Theorem 77's per-instance decision is genuinely two-sided.

The values 51, 55, 57, 62, and the $\gamma = 2\alpha$ boundary algorithm. The sweep continues past the forties. The values $51 = 3 \cdot 17$, $55 = 5 \cdot 11$ and $57 = 3 \cdot 19$ are squarefree, so the $\gamma = 2\alpha$ branch is closed by [4, Thm. 11.7], and every branch except isosceles-($\alpha+\beta$) dies arithmetically; each surviving instance is uniquely determined and the engine exhausts all six (62 dies in every branch outright, as 42 did):

instance	tile	target	nodes
N=51, M=1	$(650, 51, 676)$	$(1326, 1326, 1275)$	3649
N=51, M=3	$(70, 51, 100)$	$(510, 510, 357)$	180
N=55, M=1	$(756, 55, 784)$	$(1540, 1540, 1485)$	5911
N=55, M=5	$(24, 55, 64)$	$(440, 440, 165)$	40
N=57, M=1	$(812, 57, 841)$	$(1653, 1653, 1596)$	8961
N=57, M=3	$(88, 57, 121)$	$(627, 627, 456)$	193
N=63, M=1	$(992, 63, 1024)$	$(2016, 2016, 1953)$	15493
N=63, M=3	$(12, 7, 16)$	$(84, 84, 63)$	44859
N=63, M=7	$(8, 63, 64)$	$(504, 504, 63)$	2
N=69, M=1	$(1190, 69, 1225)$	$(2415, 2415, 2346)$	37995
N=69, M=3	$(130, 69, 169)$	$(897, 897, 690)$	523

The three N=63 rows close its isosceles-($\alpha+\beta$) branch. The value 63 itself then hangs on its $\gamma = 2\alpha$ instance — the boundary algorithm (below) leaves exactly one candidate, tile $(9, 7, 12)$ with target $(63, 63, 84)$ — and the engine exhausts it too, at 884921 nodes; to our knowledge the first value decided *inside* Beeson's "swath of ignorance" [45, 1125] for the $\gamma = 2\alpha$ branch [4, §11.6]. The values 70 and 78 die in every branch outright, as 42 and 62 did. The value 66 reduces to its $\pi/3$-equilateral instance (tile $(11, 96, 91)$, side 264, coloring number $M = 4$), which Beeson rejects [6, Thm. 7]: the tiling must witness a relation $91j = 11\ell + 96m$ with $91j < 264$, and neither $j = 1$ nor $j = 2$ admits one. And 77 goes the *other* way: its $(2\alpha, \alpha, 2\beta)$ instance satisfies Beeson's second tiling equation at $(M, s) = (5, \frac{1}{2})$ — $25 \cdot \frac{(2-\frac{1}{4})(3-\frac{1}{4})}{(1-\frac{1}{2})^2(2+\frac{1}{2})^2} = 77$ — and his four-component construction [3] realizes every solution, so *a triangle can be cut into* 77 *congruent triangles*: the $(2\alpha, \alpha, 2\beta)$ triangle over the same tile $(2, 3, 4)$ that Theorem 68 uses for 44. We need not rely on that construction: the engine finds the tiling directly (node 849 624), and its certificate is *machine-verified in Lean 4 with no axioms whatsoever* (`lean/Tiling77.lean`: all 77 triangles congruent to $16 \cdot (2, 3, 4)$ over $\mathbb{Z}[\sqrt{15}]$, contained in the target $16 \cdot (28, 16, 33)$, an explicit separating edge-line for each of the 2926 pairs, and the exact area sum). The realizability of 77 therefore rests on a kernel-checked object rather than on a cited construction.

For values that are *not* squarefree the $\gamma = 2\alpha$ branch needs more than Theorem 11.7. Beeson's isosceles paper provides exactly the needed machinery: the tile must be (k^2, m^2-k^2, mk) with $\gcd(k, m) = 1$, $2k > m > k$ [4, Lem. 11.2]; m is bounded in terms of N [4, Lem. 11.12]; the side X and base Y satisfy $X^2 = Nab$, $Y = (m/k)X$ and must decompose as $X = pa + qb + rc$ with $q, r \geq 1$ and $p + q + r < (N+1)/4$ [4, Lems. 11.11, 11.13] and $Y = ua + vb + wc$ with $v, w \geq 1$ (the base version of Lemma 11.13, whose pigeonhole proof is stated for the base); and the pair is rejected when every X-representation has $r = 1$ while every Y-representation has $w \in \{1, 2\}$ [4, Lem. 11.17]. We implemented this boundary algorithm independently (`code/gamma2alpha.py`) and calibrated it against Beeson's reported results: it excludes 20, 28 and 44 and leaves 45 surviving via the tile $(4,5,6)$, exactly as in [4, Thm. 11.18]; at 36 our filter is strictly weaker than Beeson's direct argument (it reports a surviving boundary representation), so we use it only in the sound direction. Running it on the undetermined values: *the* $\gamma = 2\alpha$ *branches at* $N = 56$ *and* $N = 60$ *are empty* — no possible boundary tiling exists.

Consequently $N = 56$ reduces far: $\gamma = 2\alpha$ is closed by the algorithm, both isosceles-$(\alpha+\beta)$ instances are exhausted by the engine (M=2: tile $(195, 56, 225)$, target $(840, 840, 728)$, 853 nodes; M=4: tile $(45, 56, 81)$, target $(504, 504, 280)$, 89 nodes), and every other branch dies arithmetically — *except* the $2\pi/3$-equilateral instance: tile $(8, 7, 13)$, equilateral target of side 56, from the factorization $(s, t) = (12, 14)$ of Proposition 64. The membership of 56 is thus equivalent to that single equilateral instance — and the engine *exhausts it*: no tiling exists, at 156646 nodes, identically in both independent engine implementations (node-for-node agreement). Likewise $N = 60$ reduces to its equilateral instance alone (tile $(5, 3, 7)$, side 30): its two isosceles-$(\alpha+\beta)$ instances are exhausted as well (M=2: tile $(56, 15, 64)$, target $(240, 240, 210)$, 123003 nodes; M=6: tile $(4, 15, 16)$, target $(120, 120, 30)$, 39 nodes) — and the equilateral instance itself is exhausted at 853731 nodes: $N = 60$ is excluded. (This also settles the m=2 member $X = 30$ of Zhang's equiconstructibility family for $(5, 3, 7)$ by exact arithmetic; Beeson had reported $X < 105$ unrealizable for this tile in unpublished floating-point work.)

Theorem 69. *No triangle can be cut into* 51*,* 55*,* 56*,* 57*,* 60*,* 62*,* 63*,* 69*, or* 78 *congruent triangles.*

Proof. The branch sweeps and the exhaustive searches above; dependencies as in Theorem 66, together with [4, Thm. 11.7] for the $\gamma = 2\alpha$ branch at the squarefree values 51, 55, 57, 69; the boundary algorithm (Lemmas 11.2–11.17 of [4], `code/gamma2alpha.py`) for the $\gamma = 2\alpha$ branch at 56 and 60; the boundary algorithm's single surviving candidate plus its 884921-node exhaustion at 63; and the 7 232 464-node exhaustion at 66 (Remark 13). □

Remark 70 (The value 70 and the unsound $g \mid M$ exclusions)**.** The value $N = 70$ is excluded. Its single surviving instance is isosceles-α: tile $(6, 5, 9)$, $M = 8$, target $(45, 45, 70)$, which passes the tiling equation *and* side-integrality ($X = 45$, and indeed $(2f + e) = 8 \mid M$); the previously published exclusion rested solely on $g \mid M$ from [3, Thm. 19] ($g = 3 \nmid 8$), whose proof is unsound (Remark 11), and Beeson's own Table 5 marks this very instance "?". The exhaustive search settles it directly: the instance admits no tiling (134 631 158 nodes, walk- and corner-pruned). The exclusion of 70 therefore rests on the search rather than on [3, Thm. 19]. The value 21, whose analogous isosceles-α candidate (tile $(2, 3, 4)$, $M = 5$, target $(12, 12, 21)$) was likewise previously killed only by the unsound $g \mid M$, *is* excluded: the engine exhausts the instance (13 659 nodes, no tiling), so Theorem 67 stands with its dependence on [3, Thm. 19] removed. The same applies to $N = 39$, whose base-β candidate (tile $(12, 7, 16)$ on $(64, 64, 117)$) the engine exhausts in 825 724 nodes: Theorem 67 therefore stands for 39 as well, independently of [3, Thm. 14]. Of the values whose exclusion rested on an unsound divisibility, only 70 remains pending (Remark 70); 59 and 66 have since been exhausted by exact search, at 1 838 175 and 7 232 464 nodes respectively.

Remark 71. The $2\pi/3$-equilateral exhaustions at $N = 56$ (tile $(8, 7, 13)$, side 56) and $N = 60$ (tile $(5, 3, 7)$, side 30) lie strictly inside the range $12 < N < 1215$ that Beeson leaves *open* for this case — he proves only $N \geq 12$ [6, §5] — and, cross-checked to the node in two independent engine implementations (156 646 and 853 731 nodes respectively), are to our knowledge the first rigorous exclusions inside it. The value 56 moreover settles the m=1 case of Zhang's equiconstructibility family for $(8, 7, 13)$: the side-56 equilateral is *not* tileable by $(8, 7, 13)$, exactly as Zhang's Conjecture 2 predicts (his construction realizes $X = 56m$ only for $m \geq 9$). It also closes the m=1 route of his Proposition 9(1) to the F_1 instance $N = 105$: any 105-tiling of $(105, 56, 91)$, if one exists, does not factor through an equilateral 56-block.

The values 76 to 80. The same branch sweep (`verify_frontier.py`) extends the determined initial segment past 75. For $N = 76 = 4 \cdot 19$ the only live branches are $\gamma = 2\alpha$ (excluded by the boundary algorithm of Lemmas 11.2–11.17 of [4], `code/gamma2alpha.py`) and the $3\alpha + 2\beta = \pi$ isosceles-(α+β) target with tile $(90, 19, 100)$ on the isosceles $(380, 380, 342)$, which the engine exhausts (955 474 nodes); every other branch is dead by its N-form. Hence 76 is not realizable. The prime $79 \equiv 3 \pmod 4$ is excluded by Theorem 1; and 77, 80 are realizable — 77 by Beeson's four-component construction [3], and $80 = 4^2 + 8^2$ by the biquadratic (commensurable) tiling [5].

Theorem 72. *No triangle can be cut into* 76 *congruent triangles. Every* $N \leq 80$ *is thereby determined, with no exception: the realizable values are exactly* 1–13*'s known members together with* 44, 77, 80 *and the standard commensurable and prime* $\equiv 1 \pmod 4$ *values, and all others are excluded. The last undetermined value below* 80 *was* $N = 70$*, settled by exhaustive search of its single surviving instance (Remark 70).*

Proof. The branch sweep and $\gamma = 2\alpha$ boundary algorithm above settle 76; Theorem 1 excludes the prime 79; Theorems 67–69 give the composite exclusions through 78 other than 70; and the four-component and biquadratic constructions realize 77 and 80. □

The values 81 to 90. The sweep continues past 80 with no change of method. Five values are realizable outright by the commensurable construction, being sums of two squares: $81 = 9^2$, $82 = 1^2 + 9^2$, $85 = 2^2 + 9^2$, $89 = 5^2 + 8^2$, $90 = 3^2 + 9^2$. The prime $83 \equiv 11 \pmod{12}$ is excluded by Theorem 2. For 86 every branch is dead by its N-form. That leaves 84, 87 and 88, at which the $\gamma = 2\alpha$ branch is alive by N-form (84 and 88 are not squarefree) and is killed by the boundary algorithm of Lemmas 11.2–11.17 of [4] (`code/gamma2alpha.py`: 82–89 all return *no possible boundary tiling*; the survivors in the band are 81 and 90, both already realizable). The surviving instances are then finitely many isosceles $3\alpha + 2\beta = \pi$ targets with a uniquely determined tile, and the engine decides each:

$$\begin{array}{lll} N = 87: & (208, 87, 256) \text{ on } (1392, 1392, 1131), & 2\,107 \text{ nodes;} \\ & (1892, 87, 1936) \text{ on } (3828, 3828, 3741), & 278\,197 \text{ nodes;} \\ N = 88: & (117, 88, 169) \text{ on } (1144, 1144, 792), & 489 \text{ nodes;} \\ & (483, 88, 529) \text{ on } (2024, 2024, 1848), & 15\,493 \text{ nodes.} \end{array}$$

All four are exhausted with no tiling. For 87 that settles the value. For 88 one further instance must be treated, from a family that cannot arise below it; see Remark 74.

Theorem 73. *No triangle can be cut into* 86 *or* 87 *congruent triangles. In the range* $81 \leq N \leq 90$ *the realizable values are exactly* 81, 82, 85, 89, 90*;* 86 *and* 87 *are excluded unconditionally;* 83 *is excluded under the hypothesis of Theorem 2 and not otherwise (Remark 76); and* 84 *and* 88 *remain open, each awaiting the exhaustion of instances currently under search.*

Proof. For 86 every branch is dead by its N-form (commensurable, tile-similar, right-angle and the four $3\alpha + 2\beta = \pi$ shapes all fail their defining equations; the $2\pi/3$ spectra of Theorems 67–69 exclude it, and the scalene $2\pi/3$ shapes cannot reach it by Remark 74; $\gamma = 2\alpha$ dies because 86 is squarefree). For 87 the same sweep leaves the two instances tabulated above, each exhausted by the exact search of Section 8. The realizable values are sums of two squares as listed, and 83 is Theorem 2. □

Remark 74 (The scalene $2\pi/3$ shape F_4 first becomes possible at exactly 88)**.** The sweeps of Theorems 67–72 test the isosceles and F_1 targets among the $2\pi/3$ shapes and omit the scalene families F_2, F_3, F_4, on the ground that their minimal counts exceed the range under consideration. That ground is exact, and it expires inside the present band. Writing $N = k^2 N_0$ with N_0 the closed-form ratio of Section 7, the minima over primitive $120°$-triples are

$$\min N_0(F_2) = 143 \text{ at } (a,b) = (3,5), \qquad \min N_0(F_3) = 264 \text{ at } (5,3), \qquad \min N_0(F_4) = 88 \text{ at } (3,5),$$

so in $N \le 90$ these three families contribute exactly one instance, and it occurs at the endpoint $N = 88$: the tile $(3,5,7)$, with its $2\pi/3$ angle opposite 7, on the scalene target $(21,55,56)$, at $k = 1$. The area ratio is exact, $660\sqrt{3} = 88 \cdot \frac{15\sqrt{3}}{2}$. It is the first instance from any of F_2, F_3, F_4 to enter the frontier sweep — scalene targets themselves are not new, the sweep of Section 7 having already searched the same tile $(3,5,7)$ on $(15,21,24)$ at $N = 24$ and $(7,8,13)$ on $(56,104,120)$ at $N = 120$ — and it is the reason 88 is not settled by the four exhaustions above. The instance is under search. The bound also confirms that no scalene $2\pi/3$ shape can affect any $N < 88$, so Theorem 72 and everything below it are untouched.

Remark 75 (The third isosceles base-α instance)**.** The isosceles base-α shape of [3, Thm. 19] must be enumerated *without* that theorem's divisibility condition $g \mid M$, whose proof is unsound (Remark 11): imposing it reports the branch empty at $N = 21$ and $N = 70$, whose instances are real and needed the engine. Solving the branch's defining cubic $N/M^2 = (1+s)(2-s^2)/\big((1-s)(2+s)^2\big)$ over the rationals in $0 < s < 1$, the solutions with $N \le 90$ are exactly three:

$$(N, M, s) = (21, 5, \tfrac{1}{2}), \qquad (70, 8, \tfrac{2}{3}), \qquad (84, 10, \tfrac{1}{2}),$$

the first two being the instances of Theorem 67 and Remark 70, exhausted at 13 659 and 134 631 158 nodes. The third is new and is the only one above 80. Since $s = e/f$ it forces $(e,f) = (1,2)$ and hence the tile $(2,3,4)$ — the same tile as the 44-tiling — on the isosceles $(24,24,42)$ with base angles α, apex at $(21, 3\sqrt{15})$. So 84 carries three instances, not the two the $(a+b)$-shape sweep produces: $(10,21,25)$ on $(210,210,84)$, already exhausted at 78 nodes, and $(110,21,121)$ on $(462,462,420)$, exhausted at 4 347 701 nodes, leaving only $(2,3,4)$ on $(24,24,42)$ under search. Sparse though this branch is — three instances in ninety values — it is not exhausted by them, which is why the band cannot be closed by N-forms alone.

Remark 76 (What 83 rests on)**.** Among the values of the band, 83 is of a different kind from the rest, and the distinction is sharper than mere corroboration. Every prime $\equiv 11 \pmod{12}$ below it — $11, 23, 47, 59, 71$ — was settled by an independent certified exhaustive search of its base-β instance, so those exclusions stand on the engine alone and need no branch theorem. For 83 the single surviving instance is the tile $(1722, 83, 1764)$ on $(3486, 3486, 3403)$, the base-β member $(e,f) = (5,6)$, and that search has not terminated. Its exclusion therefore rests entirely on Theorem 2, which is conditional on the companion's complete-corner-wall hypothesis (Remark 3). *So* 83 *is not excluded unconditionally by this paper*, and it is the first value for which that is true: everything below 81 is unconditional, either by an N-form, by Theorem 1 for primes $\equiv 7 \pmod{12}$, or by an exhaustion. We state this explicitly because the boundary between the two kinds of result is easy to lose in a table, and because the same hypothesis governs every $N \equiv 11 \pmod{12}$ above 83 as well.

Decidability. The pattern of the last two subsections is not an accident. Every branch of the classification now reduces, for each fixed N, to a *finite* list of fully determined instances: the commensurable, similar-tile, right-angle and $\gamma = 2\alpha$ branches are decided by closed N-forms; in the sporadic $2\pi/3$ shapes the divisor structure of N (Theorems 53, 55, Proposition 63) leaves finitely many (tile, scale) pairs; the equilateral cases pin the tile from finitely many factorizations (Proposition 64) or coloring numbers ([6, Thm. 2], which determines the tile from (N, M)); and the five $3\alpha + 2\beta = \pi$ shapes have finitely many (M, s) candidates by their tiling equations [3]. Each surviving instance is a bounded existence question, and the corner-anchored search of `code/engine/` decides it in finitely many steps (the branching is provably complete and the depth is N). Hence:

Theorem 77 (Decidability)**.** *It is decidable, given N, whether some triangle can be cut into N congruent triangles (under the same cited inputs as Theorem 1).*

This extends the decidability that Beeson established for equilateral targets [6, Thm. 4] to the whole problem. What decidability does *not* give is a closed-form description of the set. After Theorem 56 and Proposition 65, the necessary side is closed in *every* branch — the admissible values are Zhang's families in the sporadic shapes and explicit divisor-instances in the equilateral cases — so the entire residual openness of the problem is a single question: *which admissible instances are realized.* Zhang's trapezoid constructions realize the sporadic families for $m \geq 9$ in residue classes, and Laczkovich's elliptic-curve analysis [13] governs equilateral realizability; the open kernel is the small members that are not already realized commensurably — sharpest $N = 105$ on the F_1 target of $(8, 7, 13)$ — and each such instance is individually decidable (Theorem 77).

Remark 78 (no invariant decides realizability)**.** The directional invariants are *conserved* by the tiling process: each tile carries value $\pm(c + a - b)$ for f_α (resp. $\pm(c + b - a)$ for f_β), so placing one tile changes Φ_f of the unfilled region by $\pm V$ and the number of remaining tiles by one — in step, so that Φ_f/V stays integral and keeps the parity of the tile count throughout. Hence the only conditions such an invariant imposes are the admissibility conditions of Theorem 56, and *no* additive boundary invariant can distinguish an admissible target from a partially tiled one; none can obstruct an admissible instance, and none prunes the search below its combinatorial size. For the sharpest open instance $N = 105$ we confirmed this computationally: searching all directional weights of period ≤ 4 in the α-winding (reflections included), the only invariants are f_α ($M_\alpha = 5$) and f_β ($M_\beta = 21$), both admissible; and the vertex-type balance forced by the angle relations $a\alpha + b\beta + c\gamma \in \{\pi, 2\pi\}$ admits 42,282 nonnegative integer solutions. Deciding realizability of an admissible instance therefore lies beyond every linear or coloring invariant; it is a genuinely combinatorial question, which is why the search of Theorem 77 appears to be essential.

Remark 79 (the tiling group collapses as well)**.** The strongest standard combinatorial obstruction is the Conway–Lagarias *tiling group* $T = \langle \chi_v \rangle / \langle$ collinear additivity; tile boundary words $= 1$ in every orientation and reflection $\rangle$; a tileable region R has $\chi(\partial R) = 1$ in T. Writing edge vectors as complex numbers in $K = \mathbb{Q}(\sqrt{3}, i)$ (rotation by α is multiplication by $\zeta_\alpha = (11 + 4\sqrt{3}\,i)/13$, by $\frac{\pi}{3}$ by $(1 + \sqrt{3}\,i)/2$, reflection by conjugation) and reducing to finite quotients $\mathbb{F}_{p^2}$ with $p \equiv 11 \pmod{12}$ — so $\sqrt{3} \in \mathbb{F}_p$ but $i \notin \mathbb{F}_p$ and the plane stays genuinely two-dimensional — the tile relators, *together with* the commutators [class-1(tile), generator] that normal closure forces (a tile word's class-1 image is itself a relator, hence not central), fill the entire commutator layer. So T collapses to its rank-two abelianization (M_α, M_β): the class-2 quotient is trivial while the class-1 layer keeps rank 2. As $N = 105$ is admissible, $\chi(\partial_{105}) = 1$, and the tiling group does not obstruct it. We verified this exactly for $p \in \{23, 47, 59, 71, 83, 107, 131\}$ (`verify_tiling_group.py`), and — extending to the free nilpotent quotient through class 4 and to a range of finite non-abelian groups (through S_5, $\mathrm{GL}_2(\mathbb{F}_p)$, $\mathrm{PSL}(2, 7)$ and the relevant Frobenius groups, which carry genuine non-abelian data: sensitivity controls fire on 92–96% of tile-consistent representations) — found *no* tile-consistent quotient sending ∂_{105} off the identity, with the reptile controls $N = 4, 9$ trivial throughout and a Φ-nonadmissible loop not. The no-go of Remark 78 therefore extends past the linear invariants to the full tiling group: on this tile the tiling-group obstruction is exactly Φ, so deciding realizability of $N = 105$ lies outside every standard tiling invariant.

State of the full problem. Assembling: the commensurable branch is completely determined (squares, sums of two squares, and $2, 3, 6$ times squares, all realized [5]); the similar-tile and right-angle branches are settled in [2, 14]; the primes are settled by Theorem 51. What remains open for the full determination of the tile-count set is: sufficiency in the sporadic $2\pi/3$ branches (Zhang's completeness conjecture [9]; the admissible spectra above are the outer bounds), the $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha$ branches beyond the partial results of [3, 4], and the equilateral targets for incommensurable tiles, where Laczkovich has connected the possible N to rational points on elliptic curves [13]. With 7 and 11 excluded by [5], the primes by Theorem 51, $14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 46$ by Theorems 66 and 67, $51, 55, 56, 57, 60, 62, 63, 66, 69, 78$ by Theorem 69 (70 pending, Remark 70), 44 *realizable* by Theorem 68, 77 realized by Beeson's four-component construction [3], 28 realized by his triquadratic tilings [3], and the remaining values commensurably realized $(45, 48, 49, 50, 52, 53, 54, 58, 61, 64, 65, 68, 72, 73, 74, 75)$ or prime $(47, 53, 59, 67, 71)$, the membership of *every* $N \leq 75$ is now determined. The value $N = 76$ is likewise excluded ($\gamma = 2\alpha$ empty by the boundary algorithm; its one isosceles-$(\alpha+\beta)$ instance, tile $(90, 19, 100)$ on $(380, 380, 342)$, exhausted by the engine at 955 474 nodes); with 79 prime (Theorem 1) and $80 = 4^2 + 8^2$ realized, *every* $N \leq 80$ *except the pending* 70 is now determined (Theorem 72).

Progress on the base-β branch. Theorem 2 is conditional on Hypothesis (walls) for the branch $3\alpha + 2\beta = \pi$ at $m = 1$, and the companion paper records the following unconditional progress on that hypothesis.

For the thin subfamily $e = 1$ the hypothesis holds for $f \leq 6$, that is for the members with $N = 3f^2 - 1$ equal to $11, 26, 47, 74, 107$. For two of them the corresponding $m = 1$ target is in addition shown to admit no tiling by an independent exhaustive search (132 519 nodes at $f = 5$ and 1 251 382 at $f = 6$). These searches concern one instance each and are statements about the base-β branch at $m = 1$; they are not exclusions of the integers themselves, and indeed $74 = 5^2 + 7^2$ is a tile count by the commensurable construction.

For $e \geq 2$ the base walk is now determined. The walk of a separated member, meaning one with $f^2 > 2ef + e^2$, admits exactly three edge decompositions; the γ-trap deletes the one with no c-edge, and the remaining non-walls column $(0, e, 2e)$ is excluded outright: with no a-edge on the base each corner tile lays c on the base and a on the side, the unique fills of β, α and $\alpha + \beta$ force the corner partner and then every subsequent one, and the base is driven to consist of c-edges alone, which would require $f^2 \mid e^3$. Hence at a separated thick member the base walk is the walls form, with f a-edges, e b-edges and e c-edges. For three thick members the $m = 1$ target is likewise exhausted, establishing the hypothesis there: $(e, f) = (2, 3)$, $(2, 5)$ and $(3, 4)$, with $N = 23$, 71 and 39 and node counts 19 677, 553 417 and 825 724. The first two of these integers are primes congruent to 11 modulo 12; all three are already excluded as tile counts by other means recorded above.

A further reduction of the side condition is now available. The apex of a base-β target is rigid, and the rigidity is an identity of the whole family: $b \cos \frac{\alpha}{2} = c \cos \frac{3\alpha}{2}$ for every (e, f), which places the apex tile's third vertex exactly on the chord through the first junction of an equal side, so that its a-edge lies *along* that chord and has length exactly a. The area above that chord is $2 + b/c = N/f^2$ tile areas, filled by the two outer apex tiles together with the fraction b/c of the middle one and by nothing else, so that chord is met from above by exactly three tiles. It follows that the edge of an equal side immediately below the first junction is again a c, whence $n_c \geq 2$ and

$$pe + 2 \leq f.$$

This holds at every member with $\gcd(e, f) = 1$. The step needs only that the tile below cannot lay a b-edge towards the junction, and when $b < a$ that is forced by a semigroup argument: the residual length $a-b$ satisfies $a-b \equiv -f^2 \pmod{e}$, so any representation over $\langle a,b,c\rangle$ would need at least $e-1$ copies of b or c, giving $a-b \geq (e-1)b$, whereas $(e-1)b > a-b$ because $b > f$ holds at every member. In particular $p = 2$ is excluded on the whole tight subfamily $f = 2e+1$, by an argument independent of the chord and height estimates used there previously. The bound does not reach $p = 1$ at any new member: that would need $e \geq f-1$, and $pe+2 \leq f$ then gives $f \leq e+1 \leq pe+2-e+1$, leaving only the smallest members, all already closed.

The residual question has correspondingly sharpened. Each junction J of an equal side whose upper edge is a c determines a point $W(J)$ on the chord through J, at distance exactly a from J, namely the third vertex of the c-tile above. An a-edge immediately below J forces $W(J)$ to lie interior to an edge of some tile. Hence the hypothesis holds at any member with $e^2+ef < f^2$ provided no such T-junction occurs. At the first junction none does, because two tiles present γ there; at every later junction the configuration is a verbatim copy of the first, and a lone γ beside a straight edge is always consistent because $\alpha+\beta+\gamma = \pi$, so no counting argument can settle it. All of this is proved in the companion and the identities are machine-checked (`lean/ApexRigidity.lean`, `lean/SecondEdge.lean`).

On the tight subfamily $f = 2e+1$ the standing hypothesis is automatic — $e^2+ef < f^2$ there reduces to $e^2+3e+1 > 0$ — so $pe \leq 2e-1$ and $p \leq 1$ at every member. The members of that subfamily whose tile count $N = 11e^2+12e+3$ is a prime congruent to 11 modulo 12 and is still open are

N	e	f	tile (a,b,c)
227	4	9	$(36,65,81)$
1223	10	21	$(210,341,441)$
3011	16	33	$(528,833,1089)$
4643	20	41	$(820,1281,1681)$
5591	22	45	$(990,1541,2025)$
8963	28	57	$(1596,2465,3249)$
13127	34	69	$(2346,3605,4761)$

together with $N = 138$ at $(3,7)$, which is not prime but is the smallest member where the question is live. Each of the eight is in the same state: base walk settled, $p \in \{0,1\}$, and the single open question $p = 1$. No step of the argument uses the member beyond $e^2+ef < f^2$, so a proof at one is a proof at all eight; an exhaustive search, by contrast, reaches only the member it is run on, the next target after $N = 138$ having a triangle nine times larger in linear scale.

What remains for the hypothesis in general is the side condition, namely the exclusion of $1 \leq p \leq (f-2)/e$ where fp is the number of a-edges on a side, together with the members with $f \leq (1+\sqrt{2})e$, for which the base walk admits further columns. Both reduce to one question: whether the tile across a corner-chain b-edge lays a whole b there. At the corner itself that b-edge joins a base vertex to a side vertex and the answer is yes unconditionally; one step in, its upper end is interior, and the configurations in which an edge overruns such an end occur in genuine partial configurations at every scale examined, including $m = 1$.

Standing of the inputs. The external inputs to Theorem 1 and their publication status are as follows: Laczkovich's classification papers [11, 12] and the rep-tiling theorem of Snover–Waiveris–Williams [14] are refereed journal articles, and Beeson's isosceles paper [4] is accepted (to appear in *Integers*); Beeson's remaining branch papers [2, 3, 5, 6] and the Beeson–Zhang rationality theorem [8], which is load-bearing for the integrality of the tile, are arXiv preprints at the time of writing. Zhang's constructions [9] are cited for context only and carry no logical weight here. The parts original to this paper (Sections 3–5 and 7, together with the shape enumeration of Section 2) are self-contained given those inputs, with the arithmetic layer machine-checked (Section 8).

8. Verification

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of Section 3. `verify_invariant.py` confirms Lemma 35 ($C_f(t) = \pm(c+a-b)$ over all orientations of T), Lemma 34 (both sides agree on explicit subdivision tilings, and (6) cancels at a non-edge-to-edge incidence), and Lemma 41 (no counterexample among primitive $120°$-triples up to $c \approx 9 \times 10^6$). `verify_spectrum.py` checks Section 7: the two-positive-squares decomposition for all primes $p \not\equiv 3 \pmod 4$ below 10^5; the scale structure $b \mid k^2 \iff de \mid k$; the admissible spectrum of Theorem 53 (no prime is admissible; 33 and 46 are); and that every instance of $e \mid (a+b-c)$ with $a, b < 3000$ arises from the j-classification of Proposition 54, with none for $d = 1$. As a positive control, both necessary conditions of Theorem 42 hold on the genuine 2673-tile isosceles tiling of Herdt [4]: for its tile $(5, 3, 7)$ and scale $k = 27$, $N = k^2(a+2b)/b = 2673$ and $M = k(2b+a-2c)/(c+a-b) = -9$ are integers, as they must be on any true tiling.

The *geometric* layer is also machine-checked, end to end and with no prose joints. That the angles of the tiles meeting an interior point sum to 2π — the fact a dissection argument uses constantly and which has no counterpart in Mathlib — is obtained without any theory of angular measure, by a chain that is purely measure-theoretic: `TangentCone.poly_inter_ball_eq_coneAt` (a polytope agrees with its tangent cone at a point inside a small ball), `SectorArea.volume_sector` (the unit sector of angle θ has area $\theta/2$, via polar coordinates), `Wedge.sector_eq_halfplanes` (that polar sector is exactly the intersection of two half-planes with the unit disc), `E2Join.volume_halfplane_wed` and `E2Join.volume_wedge` (the same area statement transported to `EuclideanSpace` $\mathbb{R}$ `(Fin 2)`), and finally `AngleSumAssembled.angle_sum_interior`. Every step is axiom-clean.

The entire *arithmetic and combinatorial* layer of the proof is formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634.lean`, twenty-three theorems); every theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`:

`shape_enumeration`: the eleven-triple completeness of Section 2 — realizable corner types with $\sum m_i = 0$, $\sum k_i = 3$ are exactly the eleven listed — as pure integer arithmetic;
`k_not_dvd_sum_sub`, `M_not_int`: Lemma 41 ($(c-a-b)/\sqrt{b} \notin \mathbb{Z}$);
the full arithmetic of Theorem 42 in one master statement: no prime N satisfies the $120°$-relation, the area equation $Nb = k^2(a+2b)$, and the integrality $(c+a-b) \mid k(2b+a-2c)$ simultaneously (`iso_reduction_identity`, `prime_count_forces_scale`, `no_prime_isosceles_count`);
`add_not_prime`, `not_prime_of_two_le`, `F1_count_not_prime`–`F4_count_not_prime`: the scalene counts of Proposition 33 are composite ($a+b$ is never prime for a primitive $120°$-triple, and the F_2, F_3, F_4 counts factor);
`prime_three_mod_four_excluded`: a prime $p \equiv 3 \pmod 4$ with $p > 3$ is neither a square, a sum of two squares, nor $2n^2$, $3n^2$ or $6n^2$ (the commensurable branch of Section 2), via Fermat's two-squares theorem;
`prime_sum_two_pos_squares`: a prime $p \not\equiv 3 \pmod 4$ is a sum of two *positive* squares (the achievability half of Theorem 51);
`iso_admissible`: Theorem 53 — given $b = de^2$ with d squarefree, the area equation and Φ-divisibility force $k = dew$, $N = dw^2(a+2b)$ and $e \mid w(c-a-b)$;
`F1_product`, `F1_Ma`, `F1_Mb`, `F1_ratio`, `F1_pin`, `disc_not_square`, `tile_similar_not_prime`: the arithmetic of Proposition 37, Corollary 38 and Proposition 39 (`lean/InvariantProduct.lean`);
the finite arithmetic of the frontier sweeps, in eight theorems: the $\pi/3$-equilateral square criterion fails at 14 and 15; the first tiling equation has no admissible solution there; the two equilateral factor-pair uniqueness statements; the boundary congruences that kill

the isosceles candidates at 14 and 22; the F_1 invariant kill at 21; and the parity kill of 46.

Mathlib still has no theory of planar dissections — no `IsTiling`, no planar subdivision, no Jordan curve theorem, no Euler formula, no polytopes — so the geometric layer is built from the convexity and measure-theoretic primitives that do exist. Within that constraint it is now substantially machine-checked, and we state exactly how far, since the boundary has moved.

Formalized (`lean/Dissection.lean`). A dissection is defined as a target, N tiles, the covering equation, and pairwise disjointness of tile *interiors* — combinatorially, not measure-theoretically, so that the measure-theoretic consequence is derived rather than assumed. From it: `Dissection.aedisjoint` (disjoint interiors give a.e. disjointness, via `Convex.addHaar_frontier`; this is what lets a *non-edge-to-edge* dissection still be additive for area), the area identity `Dissection.volume_target`, and `vertex_multiplicities`, which converts a real angle equation at a vertex into the integer multiplicity equations the arithmetic consumes.

Also formalized: that each side of the target is a finite union of whole tile edges (`hasEdgeChains_edge`), by way of a support-face argument (`SupportFace.mem_convexHull_max`: a point of a simplex attaining the maximum of a linear functional is a convex combination of the vertices attaining it) together with a density step (`SegmentDense.subset_closure_diff_finite`).

Lemma 34 was the one place where geometry remained, and it is now discharged. `InvariantCore.cancellatio` derives the lemma from a single hypothesis: that the total directed length of interior tile edges is invariant under reversal of direction. That hypothesis is no longer assumed. `Dissection.g4_final` proves it for every dissection, unconditionally.

The route avoids the notion of a maximal interior segment, for which Mathlib offers no support. Instead of covering each maximal segment once from each side, one takes $S(d)$ to be the union of the interior tile edges carrying unit direction d, and shows that $S(d)$ and $S(-d)$ agree away from the finite set of tile vertices. The measure-theoretic ingredients are: a tile coincides with a half-plane near a point in the relative interior of one of its edges (`Tri.inter_ball_eq_halfplane`); a half-plane through the centre halves a ball (`volume_halfspace_inter_ball`, which Mathlib does not have); a ball inside the target is partitioned by the tiles (`Dissection.volume_ball_eq_sum` and the resulting count forces exactly two tiles at each such point (`Dissection.two_tiles_at_edge_point`), the classification of a tile's local contribution being exhaustive away from the tiles' vertices (`Tri.classify`), a finite set.

The orientation step carries the genuine geometric content, and runs as follows. The two tiles have disjoint interiors, so the open half-planes they occupy near the shared point are disjoint (`Dissection.cross_disjoint_of_onEdge`, which uses the structural disjointness of interiors rather than any measure); two nonzero linear functionals whose positive sets are disjoint are negative multiples of one another (`proportional_of_disjoint_pos`); hence the two edges span the *same* line with *opposite* senses (`Dissection.leftDir_antiparallel`); and normalising to unit directions yields $S(d)\backslash F = S(-d)\backslash F$ with F the finite vertex set (`Dissection.dirSet_horient`). Every statement named in this paragraph depends on `propext`, `Classical.choice` and `Quot.sound`, and on nothing else.

What remains on written proofs, checked numerically but not formalized, is the direction grid of Section 2 and Lemma 35. Search verdicts (`EXHAUSTED`) are declared computer assistance and are not formalized: doing so would mean verifying the search engine. Finally, a formalized theorem certifies its own Lean statement; that each such statement faithfully renders the paper result it is cited against is prose, and should be read as such.

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